Learned Transport Atlases for Exact, Gradient-Free Production MCMC

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Code, benchmark artifacts, and reproduction scripts accompany the submission.

Abstract

We present *Pathfinder Mixture-Referenced Hamiltonian Monte Carlo* (PMR-HMC), a Markov chain Monte Carlo sampler that is fast, efficient, and exact: its production phase makes *zero* calls to the target-gradient oracle and exactly *one* call to the target-density oracle per transition, while its invariant law is the target posterior for any fixed surrogate—a property that follows from the invariant-kernel construction and is additionally checked end-to-end by simulation-based calibration. Inference is split into a finite warm-up that learns geometry and a frozen production phase that exploits it. Warm-up constructs a *transport atlas*: a mixture reference $q = \sum_k w_k q_k$ whose components push a standard normal through invertible chart maps—affine Laplace charts, composed quadratic shears, conditional log-scale charts, polar (winding) charts, hierarchical location–scale charts, Student-t charts, and sinh-arcsinh triangular charts—so that each chart's Jacobian is absorbed into the exactly evaluated component density and the latent dynamics remain exactly harmonic. Production alternates chart-conditional trajectories, integrated in closed form and corrected by a single target-density Metropolis test, with independence proposals from a defensive mixture; the resulting independence kernel is uniformly ergodic whenever the proposal dominates the target. We prove exactness for arbitrary fixed cached forces, for non-injective winding charts via preimage-summed densities with Jacobian-weighted branch sampling, and for Student-t charts via an exact scale-mixture auxiliary. Under exact-atlas conditions we prove three production *complexity separations*: target-oracle cost per effective sample is $O(1)$ in the condition number κ, where gradient-based HMC requires $\Theta(\sqrt{\kappa})$ gradient calls per effective sample; $O(1)$ in the ambient dimension d at fixed residual dimension, where optimal scaling forces $\Theta(d^{1/4})$; and $O(1)$ in the separation between posterior modes, where local-flow samplers mix in time exponential in barrier height. We further give acceptance bounds separating surrogate from discretization error. Empirically, *production* target-oracle cost per effective sample is flat in condition number $\kappa \in [10^2, 10^5]$ where NUTS scales as $\Theta(\sqrt{\kappa})$, flat in dimension $d \in [50, 200]$ at fixed residual dimension, and flat in mode separation where NUTS ceases to switch modes while reporting high within-mode effective sample size. On a 37-posterior benchmark drawn from *posteriordb*, judged against gold-standard reference draws under a symmetric accuracy gate, PMR-HMC is the best of a ten-method panel—diagonal- and dense-mass NUTS, Pathfinder-initialized dense NUTS, ChEES, MCLMC, and four flow-based samplers (TESS, flow-based independence MH, a flowMC-style hybrid, NeuTra)—on 34 of 37 posteriors, with median advantages, over rows where both samplers pass the gate, of 38× over diagonal NUTS, 4.1× over dense NUTS, 10–12× over ChEES and MCLMC (which pass on only 11 and 19 of 37 rows), and 29–95× over the flow-based samplers, which fail the accuracy gate outright on the large majority of posteriors (cross-seed rank-$\widehat{R} \leq 1.05$ everywhere except the two disclosed accuracy failures); a curated 30-target controlled stress panel that deliberately retains every known failure family localizes where the method wins and loses, with every accuracy failure attributable to a named missing chart class or warm-up budget and the efficiency losses concentrated on smooth, well-scaled targets. A native C production kernel attains 0.13–385 μs per draw; end-to-end wall clock crosses over in PMR-HMC's favor beyond a median budget of $\approx 4{,}500$ effective samples.

Keywords Markov chain Monte Carlo; Hamiltonian Monte Carlo; transport maps; variational reference; Bayesian computation; surrogate modeling; exact sampling.

1 Introduction

The computational bottleneck of applied Bayesian inference is the oracle: each evaluation of the log posterior or its gradient may traverse a large data set, a differential-equation solver, or a production decision model. Gradient-based samplers in the Hamiltonian Monte Carlo (HMC) family [8, 28], and the No-U-Turn Sampler (NUTS) [13] in particular, purchase their mixing by spending gradients *continuously*: every leapfrog step of every trajectory queries ∇U, and the number of steps needed per effective sample grows with the condition number of the posterior covariance [19, 28], with dimension [3], and, without bound, with the separation between posterior modes [24].

This paper develops the converse decomposition: *first remove the structure that a quasi-Newton, optimization-driven warm-up can explain; then approximate only what remains.* A finite warm-up learns a mixture reference q built from invertible chart maps fitted to optimizer paths and pilot states. The residual

$$R(x) \;=\; U(x) + \log q(x), \qquad U = -\log \tilde{\pi}, \tag{1}$$

is then (a) nearly constant wherever the atlas fits and (b) typically low-dimensional where it does not. Production sampling requires no gradient-oracle calls at all: the Gaussian part of the latent dynamics is integrated in closed form, cached residual forces supply the correction, and a single call to the target-density oracle at the trajectory endpoint restores exactness through a Metropolis test. A calibrated independence branch mixes between modes at a rate independent of barrier height. Figure 1 summarizes the pipeline.

The central object is the *transport atlas*: a mixture $q(x) = \sum_{k=1}^{K} w_k\, q_k(x)$ whose components are push-forwards of a standard normal through chart maps T_k. Because the augmented target factorizes so that each chart's Jacobian cancels against the exactly evaluated q_k (Section 4), the latent dynamics conditional on a chart are *exactly harmonic regardless of how nonlinear the chart is.* Chart design thereby becomes a modeling language: a quadratic shear linearizes banana-shaped ridges; a conditional log-scale chart linearizes funnels; a polar chart with a wrapped-Gaussian angle circulates ring-shaped level sets; and a hierarchical location–scale chart reproduces, automatically, the non-centered reparameterization that practitioners apply by hand [4, 31]. All charts are learned from warm-up data by closed-form estimators.

1.1 Contributions

1. An exact transport-atlas kernel (Sections 4–5) whose production transitions use zero gradient-oracle calls and one target-density-oracle call, with proofs of invariance for arbitrary fixed cached forces; for non-injective (winding) charts via preimage-summed densities with Jacobian-weighted branch sampling; and for Student-t charts via an exact Gamma-auxiliary Gibbs step, developed over the full augmented space.

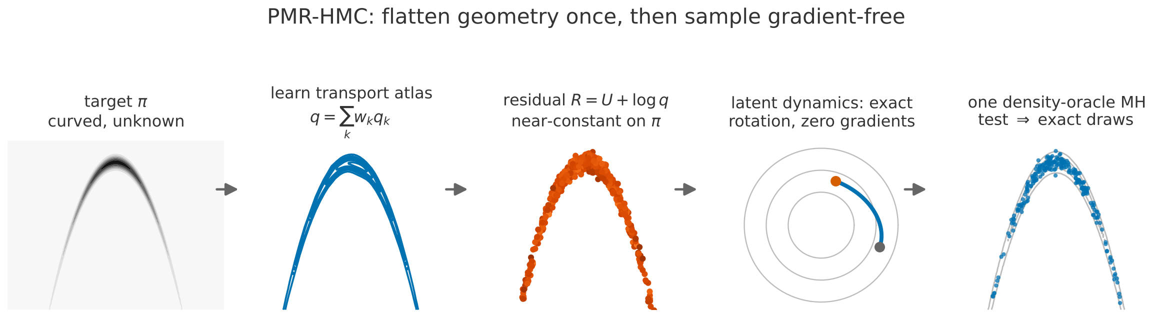

Figure 1: **The method in one picture.** A curved posterior (left) is explained by a learned transport atlas q (chart contours, blue; affine charts, vermillion); the residual $R = U + \log q$ then flattens, so the latent dynamics reduce to an exactly integrable harmonic rotation requiring no gradients, and a single target-density Metropolis test per transition restores exactness. Warm-up learns the atlas once; production never adapts.

2. Seven learned chart classes (Section 6) with consistent closed-form detection estimators, recovering analytic transports at machine precision on their canonical geometries.

3. Theory (Section 8): uniform ergodicity of the mixed kernel *conditional on* a bounded target-to-proposal ratio; acceptance bounds separating surrogate error from $O(h^2)$ discretization error; and a cost model with amortization break-even.

4. **Three complexity separations against known baseline scalings** (Theorem 4): under exact-atlas conditions we *prove* the PMR-HMC side—production target-oracle cost per effective sample $O(1)$ in condition number, $O(1)$ in ambient dimension at fixed residual dimension, and $O(1)$ in mode separation—and contrast it with the established scalings of gradient-based samplers ($\Theta(\sqrt{\kappa})$ trajectory length, $\Theta(d^{1/4})$ steps, exponential barrier mixing), which are cited, not re-proved here. Arithmetic cost is accounted separately (Table 1); the experiments show that *learned* atlases realize all three separations to close approximation.

5. A reproducible empirical study (Section 9): measured cost flat in κ, in d, and in mode separation; a 37-posterior benchmark drawn from *posteriordb* [23] with gold-reference judging and a full inclusion table, on which PMR-HMC is the accuracy-gated best of a ten-method panel—dense-mass and Pathfinder-initialized NUTS, ChEES-HMC [15], MCLMC [34], TESS, flow-based independence MH, a flowMC-style hybrid, and NeuTra—on 34 of 37 posteriors; a curated 30-target stress panel retaining every known failure family; decomposition ablations attributing the gains among atlas, dynamics, and independence branch; a production advertising model; and a native C kernel measured in both wall-clock regimes.

6. An end-to-end implementation check by simulation-based calibration [46]; exactness itself follows from the construction, and SBC serves as a pipeline audit.

2 Related Work

HMC and adaptive trajectories. HMC [8, 28] suppresses random-walk behavior via Hamiltonian flow; NUTS [13], with adaptive mass matrices [5], defines the practical state of the art. ChEES-HMC [15] replaces the recursive tree with a learned trajectory length; MEADS [16] and microcanonical samplers [34, 35] alter the underlying dynamics. All retain per-step gradient evaluation. Optimal-scaling theory gives step size $h = \Theta(d^{-1/4})$ on product targets [3] and trajectory length $\Theta(\sqrt{\kappa})$ under ill-conditioning [19, 28].

Geometry-aware and transport-preconditioned MCMC. Riemannian HMC [10] adapts to local curvature at cubic per-step cost. Split HMC [39] integrates a Gaussian component exactly; pCN and ∞-HMC [40, 41] do so against a Gaussian reference measure. Transport-map MCMC [25, 32] and NeuTra-HMC [14] precondition dynamics with learned maps; normalizing-flow-assisted samplers learn global proposals [1, 9]. Our construction differs in three respects: the reference is a *mixture of heterogeneous charts* whose Jacobians are absorbed into an exactly evaluated density; the latent dynamics remain harmonic in *every* chart; and the learned objects are deliberately low-capacity—charts fitted by closed-form regressions, with all residual error absorbed by an exact accept step rather than by model capacity.

Surrogates and delayed acceptance. Surrogate-gradient HMC [33, 45, 52], delayed-acceptance MCMC [6, 43], and learned kicks inside reversible compositions [21, 29, 44] established that approximate dynamics with exact correction preserve the target. We use this map-level exactness (Lemma 2) but approximate only $\nabla \log(\pi/q)$ in its active subspace, which is empirically far easier than the full score.

Optimization-initialized inference. Pathfinder [51] constructs Gaussian approximations along L-BFGS [22] paths; Laplace-type methods [38] exploit curvature at the mode. We extend the idea from initialization to a full sampling kernel: optimizer paths supply chart candidates, surrogate anchors, and free residual training data.

Independence samplers and ergodicity. Uniform ergodicity of the independence sampler under a bounded target-to-proposal ratio is classical [26, 36, 48]; adaptation requires care [2, 37]. Our global branch inherits the classical guarantee because adaptation is frozen; the defensive heavy-tailed component follows importance-sampling practice [12, 30].

Benchmarking. We adopt *posteriordb* [23] with its gold-standard reference draws, rank-normalized effective sample size [49], and simulation-based calibration [46].

3 Setting and Oracle Model

Let $\tilde{\pi} = e^{-U}$ be an unnormalized density on $\mathbb{R}^d$ with U continuously differentiable. The sampler may query two *target oracles*: the density oracle $x \mapsto U(x)$ and the gradient oracle $x \mapsto (U(x), \nabla U(x))$. Throughout, "one density evaluation" means one call to the target-density oracle; mixture densities, responsibilities, chart inverses, and cached forces are internal arithmetic, itemized separately in Table 1. For scalar summaries we convert a gradient call to c_g density-units and report $c_g = 2.5$ (a typical reverse-mode factor), together with raw counts and a sensitivity analysis over $c_g \in [1, 4]$ (Section 8.5); Hessian evaluations are charged d gradient calls. Efficiency is reported as target-oracle units per effective sample, using rank-normalized bulk ESS [49] minimized over coordinates; accuracy as maximum standardized moment error against analytic truth or gold reference draws.

4 The Transport-Atlas Construction

4.1 Chart-augmented target

Let $T_k : \mathbb{R}^d \to \mathbb{R}^d$, $k = 1, \dots, K$, be differentiable maps, each either a diffeomorphism or a countably-sheeted covering map (Section 5.1), and let φ denote the standard normal density. Define component densities by the push-forward

$$q_k(x) \;=\; \sum_{z \in T_k^{-1}(x)} \frac{\varphi(z)}{\bigl|\det DT_k(z)\bigr|}, \qquad q = \sum_{k=1}^{K} w_k q_k, \tag{2}$$

with weights $w \in \Delta_{K-1}$ and responsibilities $r_k(x) = w_k q_k(x)/q(x)$. When a fiber $T_k^{-1}(x)$ is countably infinite (the winding chart), the implementation truncates the branch sum at $|k_w| \leq K_w$; the discarded mass is a Gaussian tail, $\sum_{|k_w|>K_w} \varphi_1\bigl((\theta + 2\pi k_w)/c\bigr) \leq 2\,\Phi\bigl(-(2\pi K_w - \pi)/c\bigr)$, below double precision for the fitted c at K_w=64, so the truncated q_k is the density actually used—consistently in evaluation and in branch sampling—and exactness is preserved for the truncated reference.

Lemma 1 (Augmentation)**.** *$\bar\pi(x, k) \coloneqq \pi(x)\, r_k(x)$ is a probability density on $\mathbb{R}^d \times [K]$ with x-marginal π. For a diffeomorphic T_k, the conditional law of $z = T_k^{-1}(x)$ given k under $\bar\pi$ satisfies*

$$\bar\pi(z \mid k) \;\propto\; \exp\Bigl(-\tfrac{1}{2}\|z\|^2 - R\bigl(T_k(z)\bigr)\Bigr), \tag{3}$$

with R as in (1)*.*

Proof. $\sum_k r_k \equiv 1$ gives the marginal claim. For the conditional, change variables $x = T_k(z)$:

$$\begin{aligned} \bar\pi(z \mid k) &\propto \pi(T_k z)\, r_k(T_k z)\, \bigl|\det DT_k(z)\bigr| \\ &= \frac{w_k\, \pi(T_k z)}{q(T_k z)}\; q_k(T_k z)\, \bigl|\det DT_k(z)\bigr|. \end{aligned}$$

For a diffeomorphism the last factor equals $\varphi(z)$ by (2), while $\pi/q = e^{-R}/Z$; together these yield (3). □

The essential point is that the Jacobian of an arbitrarily nonlinear chart cancels against the exactly evaluated q_k: conditional on k the latent prior is exactly $\mathcal{N}(0, I)$, all non-Gaussian structure is concentrated in $R \circ T_k$, and the phase-space target $\rho_k(z, p) \propto e^{-H_k}$ with

$$H_k(z, p) = \tfrac{1}{2}\|z\|^2 + \tfrac{1}{2}\|p\|^2 + R\bigl(T_k(z)\bigr) \tag{4}$$

is dominated by an exactly solvable harmonic oscillator whenever the atlas fits: Figure 2 shows the residual over the posterior draws of a banana target spanning 22 nats under the best single Gaussian but 6 nats under the learned atlas—the flattening that cached, gradient-free dynamics require. Exact integration of a Gaussian reference part is the split-HMC mechanism [39] and, in reference-measure form, the preconditioned-Crank–Nicolson rotation of function-space samplers [40, 41]; fiber-summed densities for non-injective maps appear in surjective normalizing flows [42]. What is new here is their combination into a *learned heterogeneous mixture atlas* with Gibbs chart resampling, winding and heavy-tailed charts inside an exact kernel, and cached zero-gradient residual forces selected by the error-knee criterion.

Algorithm 1 Production transition (frozen atlas)

given state x with cached $U(x), \log q(x)$
if $\mathrm{Unif}(0,1) < p_g$ **then**
  $y \sim g$; accept w.p. (10) ▷ one target-density call
else
  $k \sim r_{\cdot}(x)$; $z \leftarrow T_k^{-1}(x)$ ▷ branch/auxiliary sampled if required
  $p \sim \mathcal{N}(0, I)$; $T \sim \mathrm{Unif}[T_-, T_+]$, $L = \mathrm{round}(T/h) \wedge L_{\max}$
  $(z', p') \leftarrow \Psi_{T/L}^{L}(z, p)$ ▷ cached forces only
  $y \leftarrow T_k(z')$; accept w.p. (8) ▷ one target-density call
end if

4.2 Production kernel

Algorithm 1 summarizes production. One local transition consists of four exactly specified maps. First, the chart is refreshed and the state lifted:

$$k \sim r_{\cdot}(x), \qquad z = T_k^{-1}(x), \qquad p \sim \mathcal{N}(0, I). \tag{5}$$

Second, the latent pair is propagated by L steps of the symmetric splitting $\Psi_h = \mathcal{K}_{h/2} \circ \mathcal{O}_h \circ \mathcal{K}_{h/2}$ with factors

$$\mathcal{O}_h(z, p) = \big(z \cos h + p \sin h,\ -z \sin h + p \cos h\big), \tag{6}$$
$$\mathcal{K}_a(z, p) = \big(z,\ p - a\, \hat{f}_k(z)\big), \tag{7}$$

the exact harmonic rotation and a momentum kick by the cached residual force. No discretization error is incurred in any direction the atlas explains. Third, the endpoint $y = T_k(z_L)$ is accepted with probability

$$\alpha_{\mathrm{loc}} = 1 \wedge \exp\big(H_k(z_0, p_0) - H_k(z_L, p_L)\big), \tag{8}$$

which makes the transition's single call to the target-density oracle (in $R(y)$); the current state's $U(x)$ and $\log q(x)$ are cached from its own acceptance. Fourth, with probability p_g the transition instead draws a global independence proposal from the defensive mixture

$$g = (1 - \epsilon)\, q + \epsilon\, t_\nu(\cdot\,; \mu_0, \Sigma_0), \tag{9}$$

accepted with the independence ratio

$$\alpha_{\mathrm{glob}} = 1 \wedge \frac{\tilde{\pi}(y)\, g(x)}{\tilde{\pi}(x)\, g(y)}. \tag{10}$$

The trajectory time is randomized, $T \sim \mathrm{Unif}[0.35\pi, 0.65\pi]$ (centered near $\pi/2$, at which the pure rotation maps $z \mapsto p$, so successive draws of an atlas-explained target are independent) to avoid harmonic resonances; the momentum flip that makes the proposal an involution (Lemma 2) is omitted from Algorithm 1 because the momentum is fully refreshed each transition. Default settings used in all experiments: $p_g = 0.2$, defensive weight $\epsilon = 0.1$ with $\nu = 4$, dual-averaging target 0.7, and step cap $L_{\max} = 48$. All quantities are frozen before production.

4.3 Cached residual forces

Two force models are precomputed per chart from warm-up anchors $\{(z_i, f_i)\}$, $f_i = J_k(z_i)^\top \nabla R(T_k z_i)$, both acting in the *active subspace*: with $C = \sum_i f_i f_i^\top$ and $V \in \mathbb{R}^{d\times s}$ its top-s eigenvectors, $\hat{f}_k(z) = V\hat{g}(V^\top z)$.

Compact scalar radial-basis energy. With Wendland kernel $\psi(r) = (1-r)_+^4(4r+1)$ [50], farthest-point centers c_j and bandwidths ℓ_j,

$$\widehat{R}_k(u) = \sum_{j=1}^{M} a_j\,\psi\left(\frac{\|u-c_j\|}{\ell_j}\right), \qquad u = V^\top z, \tag{11}$$

fitted by ridge least squares jointly on residual values and projected gradients, with closed-form gradient

$$\nabla \widehat{R}_k(u) = -20\sum_j \frac{a_j}{\ell_j^2}\left(1-\frac{\|u-c_j\|}{\ell_j}\right)_+^3 (u-c_j). \tag{12}$$

The force is conservative and vanishes smoothly outside the union of supports: the trust region is the energy's own support, so far from data the dynamics revert to exact harmonic motion—surrogate failure degrades acceptance, never correctness.

Local-affine nearest-neighbor field. Alternatively, with per-anchor Jacobians B_i from local least squares and kernel weights $\omega_i(u)$ over the m nearest anchors,

$$\hat{g}(u) = \lambda(u) \sum_{i\in\mathcal{N}_m(u)} \omega_i(u)\big[f_i + B_i(u-u_i)\big], \tag{13}$$

with a smooth distance gate $\lambda(u) \to 0$ beyond the anchor set. This field is not conservative; by Lemma 2 exactness is unaffected. The per-chart model ((11), (13), or zero force) is selected on held-out data (Section 7).

5 Exactness

Lemma 2 (Volume-preserving involution)**.** *For any fixed measurable $\hat{f}_k$, the splitting $\Psi = \mathcal{K}_{h/2}\mathcal{O}_h\mathcal{K}_{h/2}$ is volume preserving, and with momentum flip $F(z,p) = (z,-p)$ the proposal $T_{\text{prop}} = F\Psi^L$ is a volume-preserving involution.*

Proof. $\mathcal{K}_a$ is a p-shear at fixed z (unit Jacobian); $\mathcal{O}_h$ is orthogonal. From $F\mathcal{K}_aF = \mathcal{K}_a^{-1}$ and $F\mathcal{O}_hF = \mathcal{O}_h^{-1}$ we get $F\Psi F = \Psi^{-1}$, hence $(F\Psi^L)^2 = \text{id}$. Conservativeness of $\hat{f}_k$ is not required. □

Theorem 1 (Invariance)**.** *Fix the atlas, weights, caches, h, and p_g. The kernel that (i) resamples $k \sim r.(x)$, (ii) refreshes p, (iii) proposes T_{prop} and accepts with $1 \wedge e^{-\Delta H_k}$, and (iv) with probability p_g instead performs the independence step, preserves $\bar{\pi}$ and hence π.*

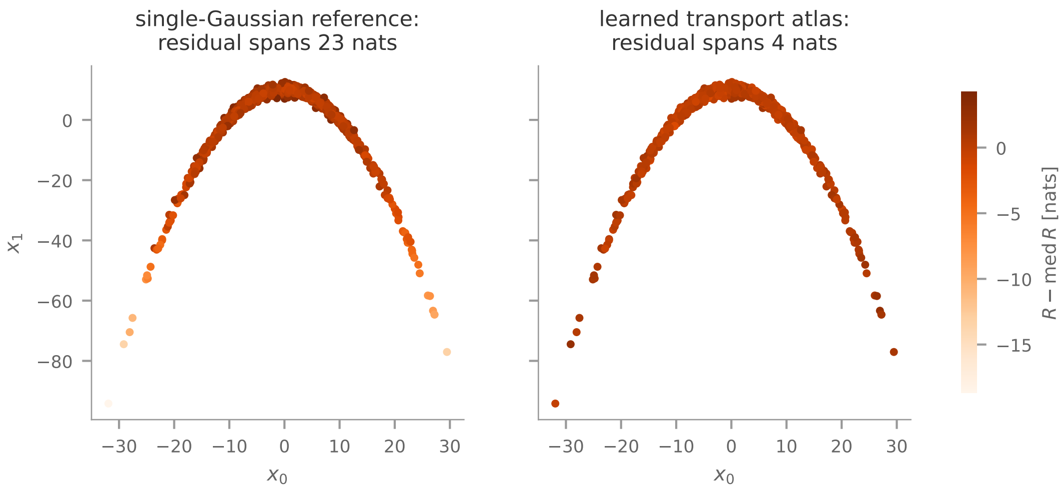

Figure 2: **What the atlas buys.** The residual $R = U + \log q$ evaluated at posterior draws of the banana target (d=20), colored on a common scale. Left: under a moment-matched single Gaussian, R varies by 22 nats along the ridge—no cached force model can track this without gradients. Right: under the learned transport atlas the same draws span 6 nats, most of it in the extreme arm tips.

Proof. Step (i) is a Gibbs update of $k \mid x$ under $\bar{\pi}$ (Lemma 1). Conditional on k, the phase-space target $\rho_k \propto e^{-H_k}$ is preserved by a Metropolis test with a volume-preserving involutive proposal [29, 48]: the flow rate

$$\rho_k(s)\left(1 \wedge \frac{\rho_k(Ts)}{\rho_k(s)}\right) = \min\{\rho_k(s),\ \rho_k(Ts)\}$$

is symmetric under $s \leftrightarrow Ts$ because $|\det DT| = 1$ and $T^2 = \mathrm{id}$. Momentum refresh preserves the Gaussian marginal; the independence step targets the x-marginal directly; a state-independent mixture of invariant kernels is invariant. □

5.1 Winding charts

The polar chart maps

$$(z_r, z_\theta) \;\mapsto\; (R + \sigma_r z_r)\big(\cos c z_\theta,\ \sin c z_\theta\big), \tag{14}$$

acting as the identity on the remaining coordinates, and is deliberately non-injective: z_θ and $z_\theta + 2\pi/c$ share an image, so a wrapped, nearly uniform angular law covers the circle and the harmonic flow in z_θ circulates it (Figure 3).

Lemma 3 (Winding exactness)**.** *Let T_k be a covering map with countable fibers and q_k defined by* (2)*. Suppose the chart inversion samples the branch $Z \in T_k^{-1}(x)$ with probability*

$$\Pr(Z = z) \;=\; \frac{\varphi(z)/|\det DT_k(z)|}{\sum_{z' \in T_k^{-1}(x)} \varphi(z')/|\det DT_k(z')|}. \tag{15}$$

Then the kernel of Theorem 1 remains $\bar{\pi}$-invariant. For charts whose Jacobian determinant is constant on each fiber—as for the polar chart (14)*, where $|\det DT_k| = \sigma_r c\,|R + \sigma_r z_r|$ depends*

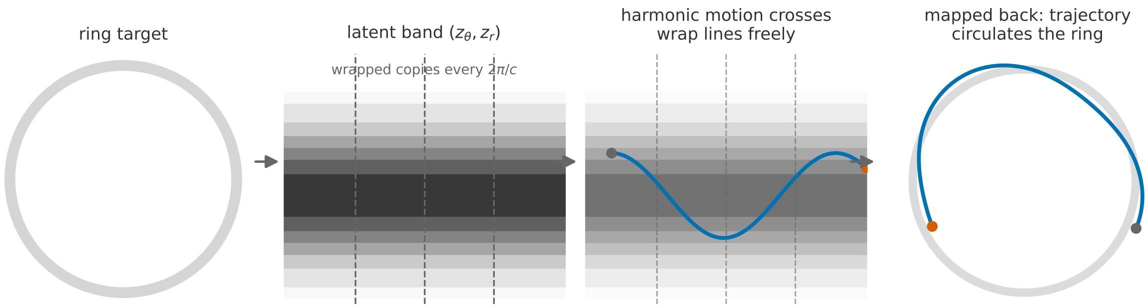

Figure 3: **The winding chart, step by step.** A closed ridge is unwrapped into a latent band on which the angular coordinate repeats every $2\pi/c$; harmonic motion travels freely along the band, and mapping back yields a trajectory that circulates the ring—the geometry that defeats arc-by-arc mixtures of local charts. Exactness under the non-injective map requires the fiber-summed density (2) and branch-sampled inversion (15) (Lemma 3).

on the fiber only through the common radius—rule (15) *reduces to sampling* $\propto \varphi(z)$*. A deterministic (principal-branch) inversion need not preserve* $\bar{\pi}$ *(and measurably breaks calibration in the ablation of Section 9.11).*

Proof. Define the latent density $\rho(z \mid k) \propto \varphi(z)\, e^{-R(T_k z)}$ on all of $\mathbb{R}^d$. Its push-forward through T_k has density

$$\sum_{z \in T_k^{-1}(x)} \frac{\varphi(z)\, e^{-R(x)}}{|\det DT_k(z)|} \;=\; q_k(x)\, e^{-R(x)} \;\propto\; \frac{q_k(x)\, \pi(x)}{q(x)},$$

the $\bar{\pi}$-conditional of x given k; and at fixed x the fiber conditional of z is proportional to $\varphi(z)/|\det DT_k(z)|$, since $e^{-R(x)}$ is fiber-constant. Sampling (15) is therefore an exact draw of $z \mid (x,k)$, and the latent Metropolis step preserves $\rho(\cdot \mid k)$ as in Theorem 1. Under a deterministic branch rule, a trajectory whose endpoint wraps is paired, on reversal, with a starting point different from the forward endpoint's latent representative, so the involution pairing of Lemma 2 fails and detailed balance with it. □

Remark 1. Equation (15) is the general statement; implementations that sample $\propto \varphi$ alone are exact only for fiber-constant Jacobians. Our polar chart satisfies this; a winding chart with fiber-varying determinant must use (15).

5.2 Student-t charts

Heavy-tailed targets defeat Gaussian atlases: π/q is unbounded and both branches stall. The scale-mixture identity

$$t_\nu(x;\mu,\Sigma) = \int_0^\infty \mathcal{N}\big(x;\mu,\Sigma/\lambda\big)\, \Gamma\big(\lambda; \tfrac{\nu}{2}, \tfrac{\nu}{2}\big)\, \mathrm{d}\lambda \tag{16}$$

restores the construction; we give the complete augmented-space argument.

Lemma 4 (t-chart exactness)**.** *Let chart k be a t-chart with parameters (μ_k, Σ_k, ν) and component density $q_k = t_\nu(\cdot; \mu_k, \Sigma_k)$ inside* (2)–(1)*. Define the augmented target on (x, k, λ) by*

$$\bar{\pi}(x, k, \lambda) = \pi(x)\, r_k(x)\, p_k(\lambda \mid x), \qquad p_k(\lambda \mid x) = \Gamma\Big(\lambda;\ \frac{\nu + d}{2},\ \frac{\nu + \delta_k(x)}{2}\Big), \tag{17}$$

with $\delta_k(x) = \|L_k^{-1}(x - \mu_k)\|^2$. Then: (i) the (x, k)-marginal of (17) *is the chart-augmented target of Lemma 1; (ii) $p_k(\lambda \mid x)$ is the exact conditional of the auxiliary in* (16)*, so drawing it is a Gibbs step; (iii) conditional on (k, λ), the change of variables $z = \sqrt{\lambda}\, L_k^{-1}(x - \mu_k)$ yields*

$$\bar{\pi}(z \mid k, \lambda) \ \propto\ \exp\Big(-\tfrac{1}{2}\|z\|^2 - R\big(T_{k,\lambda}(z)\big)\Big), \qquad T_{k,\lambda}(z) = \mu_k + \frac{L_k z}{\sqrt{\lambda}}, \tag{18}$$

with the same residual $R = U + \log q$ (the marginal-t density q_k inside q). Consequently the kernel that per transition draws $\lambda \sim p_k(\cdot \mid x)$ and then applies the local Metropolis step of Theorem 1 in the chart $T_{k,\lambda}$ preserves (17) *and hence π.*

Proof. (i) is immediate since p_k integrates to one. For (ii), Bayes' rule under (16) gives

$$p(\lambda \mid x) \ \propto\ \mathcal{N}\big(x; \mu_k, \Sigma_k/\lambda\big)\, \Gamma\big(\lambda; \tfrac{\nu}{2}, \tfrac{\nu}{2}\big) \ \propto\ \lambda^{\frac{\nu+d}{2}-1} e^{-\frac{\lambda}{2}(\nu + \delta_k(x))},$$

the stated Gamma law. For (iii), substitute $x = T_{k,\lambda}(z)$ in (17):

$$\begin{aligned}\bar{\pi}(z \mid k, \lambda) &\propto \pi(x)\, \frac{w_k q_k(x)}{q(x)}\, p_k(\lambda \mid x)\, \Big|\det \tfrac{L_k}{\sqrt{\lambda}}\Big| \\ &\propto e^{-R(x)}\, q_k(x)\, p_k(\lambda \mid x)\, \lambda^{-d/2} |L_k|.\end{aligned}$$

By (16) and (ii), $q_k(x)\, p_k(\lambda \mid x) = \mathcal{N}(x; \mu_k, \Sigma_k/\lambda)\, \Gamma(\lambda; \frac{\nu}{2}, \frac{\nu}{2})$, and $\mathcal{N}(x; \mu_k, \Sigma_k/\lambda)\, \lambda^{-d/2}|L_k| \propto \varphi(z)$ up to z-free factors; all remaining λ-dependence is z-free and cancels in ΔH. Thus (18) holds, the affine-chart argument of Theorem 1 applies conditionally on (k, λ), and the composition of the Gibbs step with the Metropolis step preserves (17). The chart-refresh step uses the *marginal* t density inside r_k, so step (i) of Theorem 1 is unchanged. □

5.3 Global branch: conditional uniform ergodicity

Theorem 2 (Uniform ergodicity under tail dominance)**.** Assume $M := \sup_x \pi(x)/g(x) < \infty$. *Then the production kernel P satisfies $P(x, \cdot) \geq (p_g/M)\, \pi(\cdot)$ and hence*

$$\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \ \leq\ \big(1 - p_g/M\big)^n \quad \textit{for all } x. \tag{19}$$

Proof. The two-case argument of Mengersen and Tweedie [26] gives the pointwise minorization

$$g(y)\, \Big(1 \wedge \frac{\pi(y)\, g(x)}{\pi(x)\, g(y)}\Big) \ \geq\ \frac{\pi(y)}{M};$$

mixing with the local kernel scales it by p_g, and (19) follows from the full-space minorization [36]. □

Remark 2 (The assumption is not automatic). The defensive t_ν component in (9) is designed to make M finite and moderate for targets whose tails are dominated by t_ν, but tail dominance is an *assumption* to be checked, not a consequence of adding a t component: a target with heavier or otherwise non-dominated tails need not satisfy it. When it holds, mode mixing costs $O(1)$ proposals per switch independent of barrier height, in contrast to the exponential mixing of local samplers on multimodal targets [24]; empirically (Figure 7c) NUTS records zero switches beyond separation four while our switch rate is flat.

6 Learned Chart Classes

Each class is fitted by a closed-form estimator (Section 7); each recovers its canonical geometry, verified as machine-precision unit tests: under the analytic funnel transport the residual spread is $\mathrm{sd}(R) \approx 10^{-15}$ with acceptance 1.000; the analytic ring chart attains acceptance 0.97.

Figure 4 shows each class acting on its canonical geometry: the defining property of every chart is that its latent view (right column) is standard normal, so the harmonic dynamics of Section 4.2 apply unchanged.

Affine (Laplace) charts. $T_k(z) = \mu_k + L_k z$, with Σ_k an |eigenvalue|-clipped inverse Hessian at evidence-selected points of L-BFGS paths [22, 51].

Composed quadratic shears. Over disjoint driver–target pairs,

$$x_{t_j} \;\mapsto\; x_{t_j} + \gamma_j\big((x_{d_j} - \mu_{d_j})^2 - m_j\big), \tag{20}$$

with unit determinant, detected by the partial-R^2 statistic (29) and peeled iteratively so that simultaneous curvatures compose into one chart. The Rosenbrock density is exactly a composed-shear Gaussian; the estimator recovers every pair coefficient ($\hat{\gamma} \in [0.999, 1.004]$, truth 1) from valley-spanning data.

Conditional log-scale charts. With driver coordinate v,

$$x_i = \mu_i + \sigma_i \exp\!\big(\tfrac{1}{2}(\alpha_i + \beta_i(v - \bar{v}))\big)\, z_i, \tag{21}$$

fitted by (30); exact for Neal's funnel [27] at $\beta \equiv 1$.

Polar (winding) charts. Equation (14), fitted by multi-candidate circle estimation with density-free angular-support gates (Section 7).

Hierarchical location–scale charts.

$$\theta_j = x_{\mathrm{loc}} + c_j\, e^{\gamma(x_v - \bar{v})}\, z_j, \tag{22}$$

whose location is a *coordinate*: the chart reproduces non-centered parameterizations [4, 31] automatically.

Student-t charts. Gaussian charts are converted by the chain-independent ray-probe test (32), with ν matched to the measured tail-growth ratio; exactness of the Gamma-auxiliary Gibbs step is Lemma 4.

Sinh-arcsinh triangular charts. A restricted Knothe–Rosenblatt transport $x_i = \mu_i + e^{\gamma_i z_{\mathrm{drv}(i)}}\, \sigma_i\, S_{\varepsilon_i,\delta_i}(z_i)$ with the Jones–Pewsey sinh-arcsinh element $S_{\varepsilon,\delta}(z) = \sinh\!\big((\operatorname{arcsinh} z + \varepsilon)/\delta\big)$: ε carries skew, δ tail weight, and an optional earlier-coordinate driver carries per-coordinate scale. Drivers are detected by a log-square regression scan; (ε, δ) by weighted-quantile matching under the pool's importance weights, admitted only when the fit beats the identity by a fixed margin (degenerate-weight fits fall back to unweighted). The inverse is

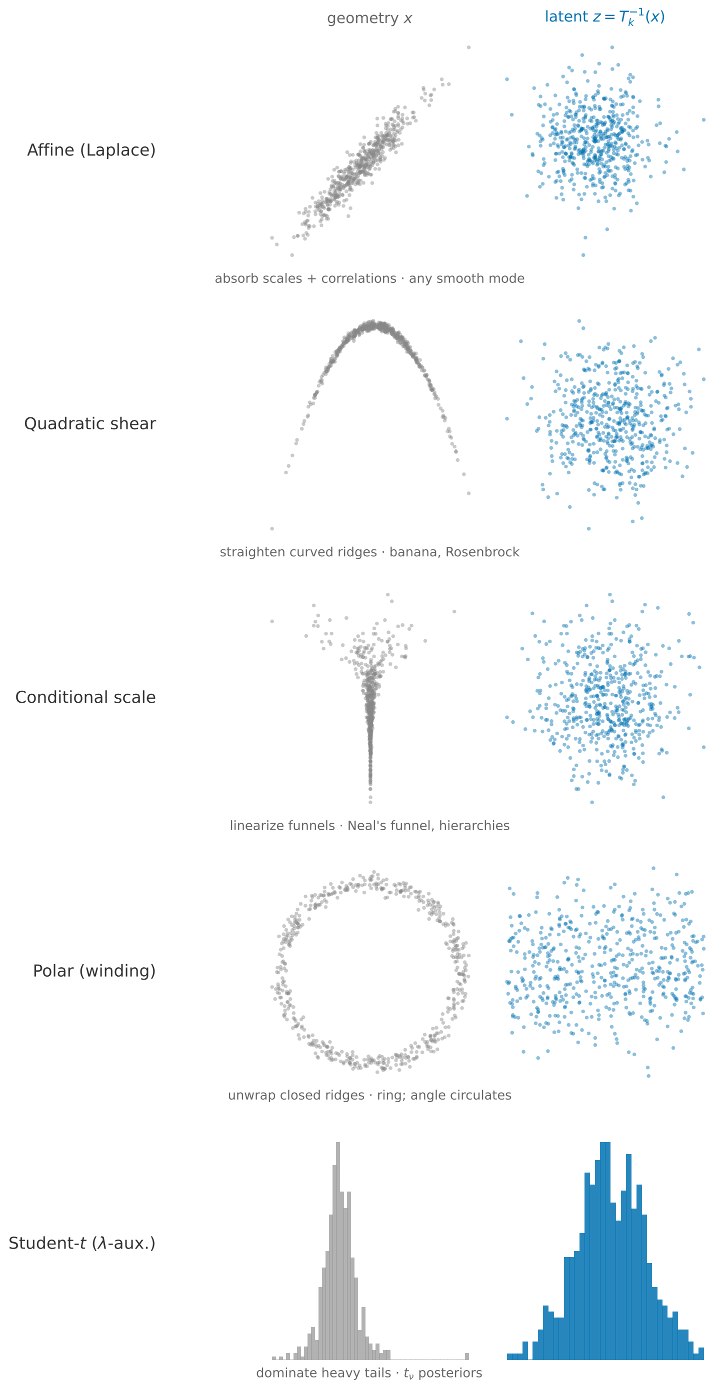

Figure 4: **Chart catalogue.** Five of the seven learned chart classes (rows; the hierarchical location–scale and sinh-arcsinh triangular classes act like the conditional-scale row with a coordinate-valued location and a skew/tail-warped marginal, respectively) each map its canonical geometry (left, gray) to an approximately standard-normal latent view (right, blue). Affine charts absorb scales and correlations; quadratic shears straighten curved ridges (banana, Rosenbrock); conditional-scale charts linearize funnels; polar charts unwrap closed ridges; Student-t charts Gaussianize heavy tails conditionally on the scale auxiliary λ. Purpose and canonical example are annotated under each row.

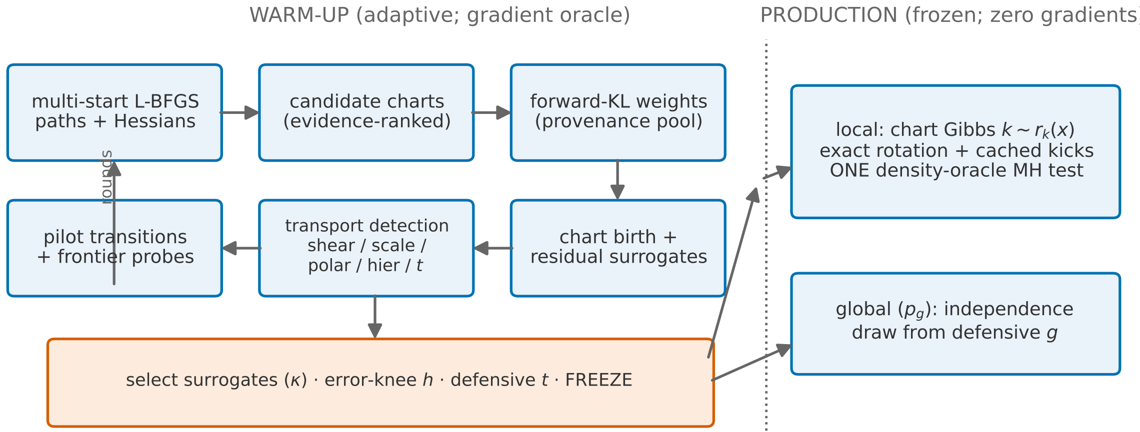

Figure 5: **Warm-up and production at a glance.** Warm-up (left of the dashed line) iterates pilot sampling, transport detection, chart birth, and weight refits, then selects surrogates, tunes the step size at the error knee, fits the defensive component, and freezes. Production (right) alternates the chart-conditional harmonic transition—one target-density call—with the global independence branch.

exact forward substitution; driver exponents are clamped with map and Jacobian sharing the clamp, so the bijection and its log-determinant remain mutually exact. Empirically this class closes the exponential-tail gap (log-cosh acceptance $0.06 \to 0.44$, now gate-passing) and helps moderate skew; extreme skew and the horseshoe remain open at the *acquisition* level—the chart is admitted but warm-up data does not yet expose enough driver variation to fit it well (Section 10).

7 Warm-Up Estimators

Every estimator is closed-form, so the atlas is a deterministic function of warm-up data; Algorithm 2 gives the loop and Figure 5 the corresponding flowchart.

7.1 Candidate charts and evidence scores

Multi-start L-BFGS paths supply candidate centers μ; covariances are clipped inverse Hessians

$$\Sigma = V \operatorname{diag}\big(\max(|\lambda_i|, \lambda_{\min})\big)^{-1} V^\top, \qquad H = V\Lambda V^\top, \tag{23}$$

ranked by the Monte Carlo evidence bound

$$\mathcal{E}_k = \frac{1}{S}\sum_{s=1}^{S}\big[-U(\mu_k + L_k z_s)\big] + \frac{1}{2}\log|\Sigma_k|, \qquad z_s \sim \mathcal{N}(0, I), \tag{24}$$

and deduplicated by Mahalanobis distance under retained charts.

7.2 Mixture weights: forward KL on a provenance pool

Weights maximize the concave pooled objective

$$w^\star = \arg\max_{w \in \Delta_{K-1}} \sum_i \bar{\omega}_i \log\Big(\sum_k w_k q_k(x_i)\Big) \tag{25}$$

via the EM fixed point $w_k \leftarrow \sum_i \bar{\omega}_i\, r_k^w(x_i)$, where the pool weights are truncated self-normalized importance ratios [17]

$$\bar{\omega}_i \;\propto\; \min\Big\{\frac{\tilde{\pi}(x_i)}{m(x_i)},\; \sqrt{N}\cdot\overline{\tilde{\pi}/m}\Big\}, \tag{26}$$

computed against the pool's *generating* mixture m: the balanced mixture of every component that ever contributed pool samples, including since-pruned ones. Substituting the current mixture for m is a proposal mismatch that we observed to collapse the fit (importance ESS $\to 1$) and delete correct charts; (26) is a correctness-of-fit detail, not a tuning choice.

7.3 Surrogate selection

Per chart, candidate force models are compared on held-out anchors by the normalized force error

$$\kappa = \frac{\sum_{i\in\mathrm{val}} \|f_i - \hat{f}(z_i)\|^2}{\sum_{i\in\mathrm{val}} \|f_i\|^2}, \tag{27}$$

whose denominator is the zero-force baseline: a model is deployed only if $\kappa < 1-\varepsilon$, with a cost-aware tie-break toward (11). This rule lets a chart switch its cache *off* on geometry it cannot represent rather than degrade below exact harmonic motion.

7.4 Residual-guided chart birth

With coverage deficit $a(x) = \log\tilde{\pi}(x) - \log q(x)$, frontier states maximize

$$F(x) = \big(a(x) - \operatorname{med} a\big)_+ \big(1 - \operatorname{trust}(x)\big) \tag{28}$$

over pilot states and overdispersed chart probes; promising probes are refined by a capped ($\leq$ 4-step) descent—enough to relax the off-manifold error of tangent probes on curved ridges, while an uncapped descent collapses candidates back to the dominant mode. Newborn charts take clipped-Hessian covariances (23) with a geometry-uncertainty widening, and weights are refit by (25).

7.5 Transport detection

Composed shears. For a driver–target pair (d,t), let $\tilde{q}$ be the component of $(x_d - \bar{x}_d)^2$ orthogonal to $\operatorname{span}\{1, x_d\}$ over the fit set. The pair is admitted when the partial coefficient of determination

$$R^2_{d\to t} = \frac{\langle \tilde{q},\, x_t\rangle^2}{\|\tilde{q}\|^2\,\mathrm{Var}(x_t)} \tag{29}$$

exceeds a threshold; $\gamma = \langle \tilde{q}, x_t\rangle / \|\tilde{q}\|^2$, the fitted shear is peeled from the data, and the scan repeats. Plain R^2 is deflated by the linear leakage of $(x_d - \bar{x}_d)^2$ under skewed data; (29) is the skew-robust statistic.

Conditional scale. For driver v, slopes follow from the bias-corrected log-square regression

$$\log(x_i - \mu_i)^2 \;\approx\; \alpha_i + \beta_i(v - \bar{v}) + \mathbb{E}\log\chi_1^2, \qquad \mathbb{E}\log\chi_1^2 \approx -1.2704. \tag{30}$$

Polar. Circle candidates come from the data mean, the Kåsa linear system [18]

$$\begin{bmatrix} 2x & 2y & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ t \end{bmatrix} = x^2 + y^2, \tag{31}$$

and variance-of-radius descent from each; a candidate survives only under density-free *angular-support* gates (sector occupancy $\geq 10/24$; maximal angular gap $\leq \pi$), which reject the tangent-circle degeneracy of short-arc fits, plus a held-out density comparison against a Gaussian alternative.

Hierarchical location–scale. A (driver, location) pair scan applies (30) to deviations $x_j - x_{\mathrm{loc}}$; the shared exponent is $\gamma = \bar{\beta}/2$.

Heavy tails. Along latent rays $z = ru$ of the dominant chart,

$$\rho = \frac{U(T(r_3u)) - U(T(r_2u))}{U(T(r_2u)) - U(T(r_1u))}, \qquad (r_1, r_2, r_3) = (3, 9, 27), \tag{32}$$

equals $(r_3^2 - r_2^2)/(r_2^2 - r_1^2) = 9$ for Gaussian tails and is $O(1)$ for polynomial tails; a measured $\rho < 4$ triggers t-conversion with ν matched to ρ. Ray probes are chain-independent, avoiding the circularity of estimating tail weight from a chain confined to the bulk.

7.6 Step size: dual averaging and the error knee

Dual averaging [13] targets endpoint acceptance; a final refinement fits the two-term error model of Proposition 1 from matched trajectory batches at h and $h/2$:

$$\widehat{V}_{\mathrm{sur}} = \frac{16V(h/2) - V(h)}{15}, \qquad \widehat{ch^4} = \frac{16}{15}\big(V(h) - V(h/2)\big), \tag{33}$$

with $V(h) = \mathrm{Var}(\Delta H)$, moving h to the knee $ch^4 \approx V_{\mathrm{sur}}$: below it, smaller steps buy no acceptance because surrogate error is h-independent.

8 Theoretical Analysis

8.1 Acceptance bounds

Proposition 1 (Error decomposition)**.** *Let $e_k(z) = \nabla_z R(T_k z) - \hat{f}_k(z)$ and suppose $\|e_k\| \leq \delta$ along a trajectory of total time T with $\mathcal{L}_p = \int_0^T \|p_t\|\,\mathrm{d}t$, with $R \circ T_k$ and $\hat{f}_k$ of class C^2 on its neighborhood. Then*

$$|\Delta H_k| \;\leq\; \delta\,\mathcal{L}_p + C_T h^2, \qquad \alpha_{\mathrm{loc}} \;\geq\; e^{-\delta\mathcal{L}_p - C_T h^2}. \tag{34}$$

If moreover $\hat{f}_k = \nabla \widehat{R}_k$ with $\sup|R \circ T_k - \widehat{R}_k - c| \leq \eta$, then $|\Delta H_k| \leq 2\eta + C_T h^2$.

Algorithm 2 Warm-up (all adaptation frozen before production)

1: multi-start L-BFGS; Hessian charts at evidence-selected path points (23)–(24); collect $(x, U, \nabla U)$ anchors
2: fit weights (25)–(26); prune by leave-one-out coverage
3: bootstrap transports from path anchors; t-conversion by (32)
4: **for** round $= 1, \ldots, B$ **do**
5: pilot transitions (states, anchors, frontier candidates)
6: retry transport detection on all geometry points; birth charts by (28); refit weights; rebuild caches
7: **end for**
8: select surrogates by (27); dual-average h; error-knee refinement (33); fit defensive component of (9); **freeze**

Proof. Along the continuous surrogate flow $\dot{H}_k = p^\top e_k(z)$, giving the first term by Cauchy–Schwarz; the $O(h^2)$ term follows, for conservative $\hat{f}_k$, from the backward-error bound for symmetric splittings of the surrogate Hamiltonian [20], and for non-conservative fields from the global $O(h^2)$ trajectory-error bound of the second-order splitting over the fixed horizon T. For the scalar case write $H_k = \widehat{H}_k + (R \circ T_k - \widehat{R}_k)$; the splitting nearly conserves $\widehat{H}_k$ and the difference is bounded by 2η end-to-end. □

8.2 Dimension-independent target-oracle efficiency

Theorem 3 (Target-oracle dimension independence)**.** *Suppose that after an atlas chart, $\pi = \pi_s \otimes \mathcal{N}(0, I_{d-s})$ with q matching the Gaussian factor exactly, charts acting block-diagonally, and cached forces supported on the first s coordinates (as induced by the active-subspace construction). Then ΔH_k is a function of the s-block alone, so acceptance, the tuned (h, T), and the number of* target-oracle *calls per effective sample of s-block statistics are independent of d.*

Proof. On the Gaussian block R is constant and $\hat{f}_k$ vanishes, so the dynamics there are the exact rotation (6): energy is conserved exactly, coordinate-wise, and the block contributes zero to ΔH_k deterministically. All acceptance statistics coincide with those of the s-dimensional sampler, and each transition makes one density-oracle call regardless of d. □

Remark 3 (What is, and is not, dimension-independent). Theorem 3 concerns *target-oracle* efficiency. Per-transition *arithmetic* includes $O(Kd^2)$ mixture algebra (Table 1), and the cost of a single target evaluation may itself grow with d; neither is claimed independent of dimension. In the oracle-dominated regimes this paper targets, the oracle count is the relevant complexity; wall-clock results (Section 9.9) quantify the arithmetic explicitly.

Corollary 1 (Conditioning)**.** *For Gaussian targets with arbitrary covariance, an exact Laplace chart gives $R \equiv$ const, acceptance one, and $O(1)$ target-oracle units per (near-independent) effective sample, independent of κ; the $\Theta(\sqrt{\kappa})$ dependence is confined to the one-time L-BFGS warm-up.*

8.3 Three complexity separations

The preceding results combine into the paper's central theoretical statement: along the three axes on which gradient-based samplers are known to degrade, the *production* cost of PMR-

HMC does not degrade at all—the entire dependence is confined to the one-time warm-up.

Theorem 4 (Complexity separations)**.** *Every production transition makes exactly one target-density-oracle call and zero gradient-oracle calls. Moreover:*

(i) ***Conditioning.*** *For a Gaussian target with condition number κ and an exact Laplace chart, trajectories at $T = \pi/2$ are accepted with probability one and successive states are independent draws from π: $O(1)$ target-oracle units per effective sample, uniformly in κ. Leapfrog-based HMC requires $\Theta(\sqrt{\kappa})$ gradient calls per effective sample [19, 28].*

(ii) ***Dimension.*** *Under the hypotheses of Theorem 3, target-oracle units per effective sample of the s-block statistics are independent of the ambient dimension d. Optimal scaling for leapfrog HMC on product targets forces step size $h = \Theta(d^{-1/4})$ and hence $\Theta(d^{1/4})$ gradient calls per effective sample [3].*

(iii) ***Multimodality.*** *Under the tail-dominance hypothesis of Theorem 2, for any measurable set A the expected number of transitions to first entry of A is at most $M/(p_g\,\pi(A))$,* independent of the separation between modes*; local-flow samplers have mixing times exponential in barrier height [24].*

Proof. The per-transition oracle count is Algorithm 1. (i) With an exact chart, $R \equiv$ const, so $\Delta H_k = 0$ deterministically and every proposal is accepted; at $T = \pi/2$ the rotation (6) maps $(z, p) \mapsto (p, -z)$, and since p is refreshed i.i.d. $\mathcal{N}(0, I)$ each transition, successive positions are independent draws from the latent Gaussian, hence from π (Lemma 1). (ii) is Theorem 3. (iii) By the minorization in the proof of Theorem 2, $P(x, A) \geq (p_g/M)\,\pi(A)$ for every x, so the first-entry time of A is stochastically dominated by a geometric random variable with success probability $(p_g/M)\,\pi(A)$, giving the stated expectation bound. No quantity in the bound depends on the location or separation of the regions carrying π's mass. □

Remark 4 (Idealization and its empirical gap). All three statements are exact-atlas idealizations; with *learned* atlases the experiments measure their realization directly (medians over 8 seeds): production cost 1.26–1.35 units/ESS across $\kappa = 10^2..10^5$ (slope 0.00 against the $\sqrt{\kappa}$ reference), a 24–37 units/ESS plateau across $d = 50..200$, and a mode-switch rate flat in separation while NUTS's median falls to zero (Figure 7). The warm-up, not production, absorbs the $\Theta(\sqrt{\kappa})$ optimization cost and the mode-discovery problem; Section 10 discusses when warm-up itself fails.

8.4 Cost model and amortization

Table 1 itemizes each phase. Writing G for total warm-up gradient calls and N for production draws, per-draw target-oracle costs compare as

$$\underbrace{c_g G/N + 1}_{\text{PMR-HMC, amortized}} \qquad \text{vs.} \qquad \underbrace{c_g L_N}_{\text{NUTS}}, \tag{35}$$

with break-even draw count

$$N^\star \;\approx\; \frac{c_g G}{c_g L_N - 1}, \tag{36}$$

a few hundred draws in every experiment below. G is paid once per posterior while L_N is paid per draw indefinitely; chains share one frozen atlas, dividing G further.

Table 1: Oracle and arithmetic cost by phase (n_H Hessians; A anchors; L integrator steps; M radial-basis centers; s active dimension).

phase	target-oracle cost	arithmetic
L-BFGS starts	c_g·iters	$O(d)$/iter
Hessian charts	$c_g\, n_H d$	$O(d^3)$/chart
anchors, probes, pools	$c_g A + O(A)$	$O(Ad^2)$
weight fit (25)	—	$O(KN_{\text{pool}}d^2)$
surrogate fits	—	$O(AMs + Ms^3)$
tuning and knee (33)	$O(10^3)$ density	—
production, local	1 density	$O(Kd^2 + LMs)$
production, global	1 density	$O(Kd^2)$
NUTS, per draw	$c_g L_N$	$O(L_N d)$

8.5 Oracle accounting and sensitivity

All summary efficiencies use $c_g = 2.5$; raw density and gradient counts are released with the artifacts. Because production uses zero gradient calls, the PMR-HMC-to-NUTS efficiency ratio is monotone in c_g: with per-draw costs (35) it scales as

$$\text{ratio}(c_g) = \frac{c_g L_N}{c_g G/N + 1}, \tag{37}$$

so moving c_g from 2.5 to 1 multiplies reported ratios by a factor in $[0.4, 1]$ and to 4 by a factor in $[1, 1.6]$. On the ill-conditioned Gaussian, recomputed from the 8-seed sweep with the warm-up's density/gradient mix bounded both ways, the amortized κ=10^4 ratio lies in roughly $[143, 223]\times$ at c_g=1 and $[264, 358]\times$ at c_g=4, around the reported 251× at c_g=2.5. No qualitative conclusion in Section 9 changes over $c_g \in [1, 4]$.

9 Experiments

9.1 Protocol

Every method runs on the same two final suites: the 37-posterior posteriordb battery (gold-judged) and a curated 30-target controlled stress panel (truth/reference-judged). The ten-method panel comprises PMR-HMC, diagonal-mass NUTS, and dense-mass NUTS (8 seeds on posteriordb, 4 on the stress panel), plus Pathfinder-initialized dense-mass NUTS, ChEES-HMC, MCLMC, TESS, flow-based independence MH, a flowMC-style flow-MALA hybrid, and NeuTra-HMC (4 seeds; the four flow-based methods share one RealNVP architecture and training budget, charged to their oracle counts). The three scaling sweeps use 8 seeds. ESS is rank-normalized bulk ESS on a single chain, minimized over coordinates; cross-seed rank-$\widehat{R}$ diagnostics accompany the posteriordb battery. Accuracy is judged against analytic truth or gold reference draws under one symmetric gate (max standardized mean error $\leq 0.25\sigma$ and max log-variance error ≤ 0.5); all numbers, raw oracle counts, and scripts are released.

Per-method draw budgets and tuning, disclosed in full: PMR-HMC draws 8,000 production samples per posteriordb chain and 6,000 on the stress panel; NUTS variants 2,500 after 700–800 warm-up iterations; ChEES 1,500 per chain over 16 chains after 500 adaptation steps

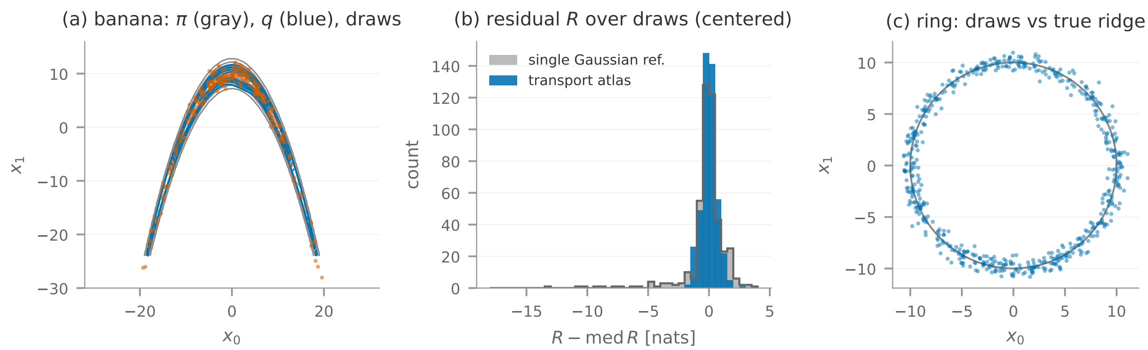

Figure 6: Mechanics. (a) The learned atlas (blue: contours of q with a fitted shear chart) reproduces the banana target (gray); production draws (vermillion) cover both arms. (b) The core of the construction: the residual $R = U + \log q$ under the transport atlas (blue) versus under a moment-matched single Gaussian (gray)—the atlas flattens R from a $\sim$22-nat range to a few nats, which is what zero-gradient cached dynamics require. (c) Visual check on the ring: draws against the true ridge; the winding chart's harmonic angle circulates the full circle.

(identity mass, Adam 0.02, per-chain oracle accounting); MCLMC 8,000 after the standard 2,000-step L/step-size auto-tune [34]; the flow-based methods 6,000 (2,500 for NeuTra) after a shared 800-step RealNVP training run charged to their oracle counts. No covariates are standardized for any method. Because the headline metric is *amortized* (warm-up included), a longer production chain dilutes warm-up more for PMR-HMC than for the fast-warm-up baselines; the scaling study reports production-only and amortized rates separately for exactly this reason, and the crossover analysis of Section 9.9 quantifies the budget dependence in wall clock.

9.2 Complexity separations

Figure 7 is the measured realization of Theorem 4: the NUTS median grows from 153 to 5,672 units/ESS over $\kappa = 10^2..10^5$ (fitted slope 0.52) with tight interquartile ranges, while PMR-HMC production is flat at 1.26–1.35 (slope 0.00), i.e., 365× amortized and 3,500× production-to-production at $\kappa = 10^5$. Embedded bananas: PMR-HMC production plateaus at 24–37 units/ESS across $d = 50..200$, while NUTS medians range 1,013–3,547 with interquartile ranges reaching 2.5×10^4—on curved targets NUTS is not merely expensive but strongly seed-sensitive. Mixtures: the NUTS median switch count falls $42 \to 1 \to 0 \to 0$ across separations 3..8 with median error rising to 0.65σ while it reports high within-mode ESS; PMR-HMC holds a median ≈ 570 switches and $\leq 0.04\sigma$ error at every separation, with interquartile ranges of a few tens of switches. Figure 6 shows the underlying mechanics on the canonical geometries (atlas fit, residual flattening, winding coverage), and Figure 8 the corresponding bimodal traces, convergence-versus-oracle-cost curves, and funnel draws.

9.3 A 37-posterior benchmark drawn from posteriordb, against ten methods

The posteriordb battery runs 37 ported posteriors (including the arK, ARMA(1,1), and GARCH(1,1) time-series models with their constraint Jacobians) against a ten-method panel

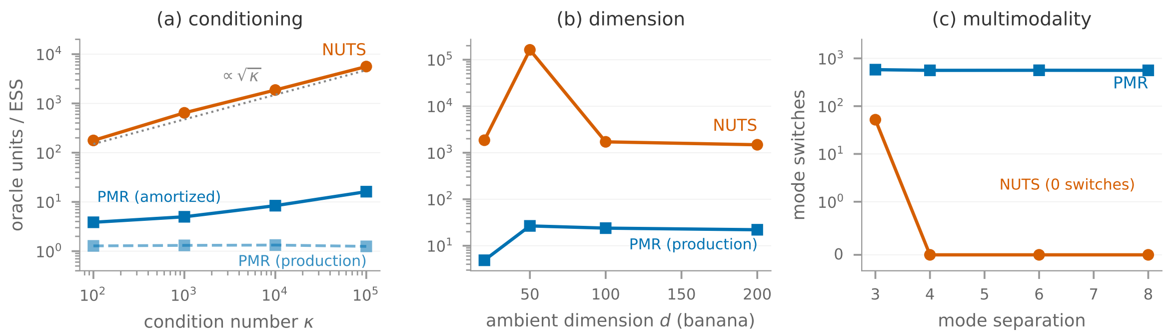

Figure 7: Measured separations (target-oracle units per effective sample; medians with interquartile bands over 8 seeds). (a) NUTS follows the $\sqrt{\kappa}$ slope over $\kappa = 10^2..10^5$ while PMR-HMC production cost is flat at 1.26–1.35 (Corollary 1); the solid blue curve includes amortized warm-up. (b) On embedded bananas PMR-HMC production cost plateaus while NUTS is both expensive and strongly seed-sensitive (the wide band spans its pathological runs). (c) Mode switches versus separation: the NUTS median falls to zero while its within-mode ESS stays high; the global branch is flat (Theorem 2).

judged on the database's gold reference draws: PMR-HMC, diagonal-mass NUTS, and dense-mass NUTS at 8 seeds, plus Pathfinder-initialized dense NUTS, ChEES-HMC, MCLMC, and four learned-transport samplers—TESS, flow-based independence MH, a flowMC-style flow-MALA hybrid, and NeuTra-HMC, all four sharing one RealNVP training budget—at 4 seeds. Cross-seed rank-$\widehat{R}$ is computed per target. Table 3 is the inclusion audit (37 of 46 gold-referenced posteriors; the remainder require ODE, HMM-forward, or latent-GP infrastructure—no posterior was excluded on account of results). Under the *accuracy-gated* best criterion applied symmetrically to all samplers (a row is contestable only when a sampler's maximum gold error is below 0.25σ mean and 0.5 log-variance; a violation is a failure for the violating sampler regardless of its efficiency), **PMR-HMC is the gated-best method on 34 of 37 posteriors against all nine baselines simultaneously** (dense-mass NUTS takes 2, MCLMC takes 1). We report efficiency ratios over rows where *both* PMR-HMC and the baseline pass the gate—a gate-failing chain's ESS is not a meaningful denominator, by the paper's own argument—and give each baseline's gate-fail count alongside. The median PMR-HMC advantage is 38× over diagonal NUTS (35 contested rows), 4.1× over dense-mass NUTS (35; 90th percentile 44×), 4.0× over Pathfinder-initialized dense NUTS, 12× over ChEES (which passes the gate on only 11 of 37 rows), 10× over MCLMC (19 rows), 95× over NeuTra (31), and 11–45× over TESS, flow-IMH, and the flowMC-style hybrid, which pass the gate on at most 9 of 37 posteriors each—their dominant failure mode on real posteriors is accuracy, not speed (Section 9.4). Dense mass adaptation shrinks the margins exactly as expected—it is the strongest baseline—but flips only two verdicts. Table 2 shows representative rows; the complete 37 × 10 table is Table 7 (Appendix A). The three PMR losses are informative and doubly flagged: the two accuracy failures (`earn_height`, an extreme heteroscedastic raw-scale tail, and `kilpisjarvi`) are precisely the two targets whose cross-seed rank-$\widehat{R}$ exceeds 1.05 (1.11 and 1.26; all other targets ≤ 1.01), corroborating the gold-error verdicts; the one efficiency loss is noncentered eight-schools, where MCLMC is best (43.6 vs. 8.1)—a model the practitioner has already reparameterized is the gradient samplers' best case. Figure 9 visualizes ratios and

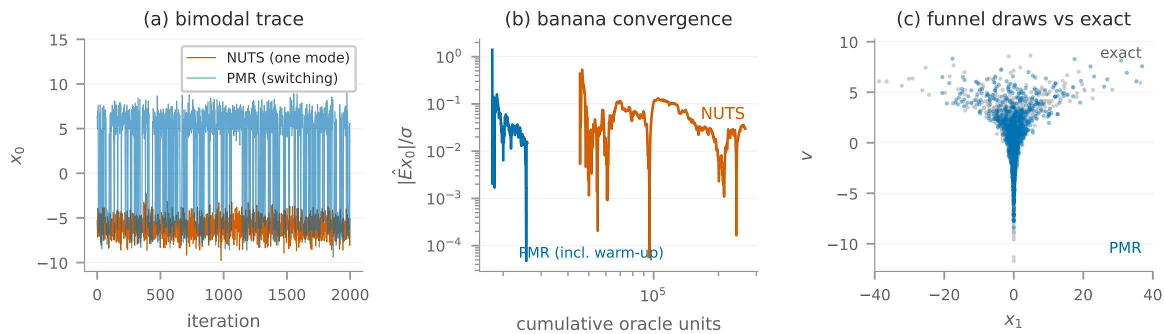

Figure 8: Progression and visual checks. (a) Bimodal traces: NUTS remains in one mode; PMR-HMC switches freely. (b) Running estimate error versus cumulative target-oracle units on the banana (log–log): the PMR-HMC curve is shifted by warm-up, then converges far to the left of NUTS. (c) Funnel draws (blue) against exact samples (gray): the learned conditional-scale chart reaches both mouth and neck.

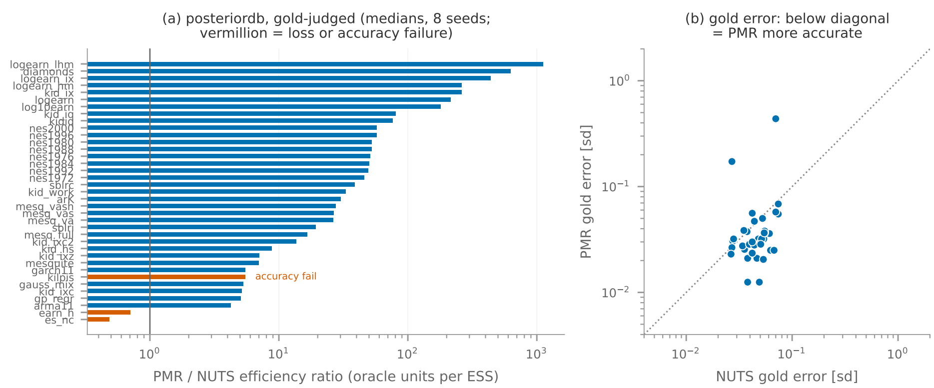

Figure 9: posteriordb summary. (a) Efficiency ratio per posterior (log scale). (b) Accuracy versus gold draws: points below the diagonal are posteriors where PMR-HMC is more accurate than NUTS.

accuracy.

9.4 The ten-method panel: gradient samplers and learned transports

Table 4 summarizes the panel across both suites; the complete per-target tables are Appendix A. Three findings organize the comparison. *First*, dense-mass NUTS is the strongest baseline overall: it improves diagonal NUTS by an order of magnitude on correlated targets, never fails the gate on posteriordb, and is the runner-up by median efficiency ratio on both suites (by gated-best count the mega-30 runner-up is MCLMC), yet PMR-HMC retains a 4.1× (2.1×) both-pass median advantage on posteriordb (mega-30), and Pathfinder initialization changes its totals only marginally. *Second*, MCLMC is the strongest baseline on *smooth, unimodal, well-scaled* targets: on the controlled mega-30 families it is gated-best on 9 of 30 rows (GLM-type regressions, log-cosh, moderate skew—including rows where PMR's chart

Table 2: Representative posteriordb rows, gold-judged, with the full ten-method panel (min-ESS per 10^3 target-oracle units; PMR/NUTS/dNUTS medians over 8 seeds, other methods over 4; * fails the accuracy gate; **bold** = best among gate-passing methods. All 37 rows: Table 7.)

posterior	d	PMR	NUTS	dNUTS	PF-dN	ChEES	MCLMC	TESS	f-IMH	flowMC	NeuTra
logearn_lhm	4	**206.3**	0.2	23.7	21.9	0.0^*	0.1^*	0.3^*	0.2^*	25.1^*	0.1^*
diamonds	26	**63.2**	0.1	14.7	10.1	0.0^*	0.1^*	0.0^*	0.0^*	0.0^*	0.0
logearn_ix	5	**193.6**	0.4	3.7	2.6	6.4^*	0.1^*	0.0^*	0.0^*	0.0^*	0.1
logearn_hm	4	**219.1**	0.8	4.5	3.2	12.7^*	0.1^*	0.0^*	0.0^*	0.0^*	0.5
kid_ix	5	**155.5**	0.6	20.4	15.3	15.2^*	0.1^*	0.0^*	0.0^*	0.1^*	0.1^*
logearn	3	**239.0**	1.1	4.1	2.6	19.2^*	0.1^*	0.0^*	0.0^*	50.1^*	0.8
log10earn	3	**214.5**	1.2	1.9	1.4	23.1^*	0.1^*	0.0^*	0.0^*	0.0^*	1.3
kid_iq	3	**228.1**	2.8	25.9	17.1	18.4^*	0.2^*	0.0^*	0.0^*	0.0^*	1.6
kid_hs	3	**84.4**	9.6	37.6	33.1	96.3^*	0.5^*	0.0^*	0.0^*	0.0^*	2.5
kilpis	3	0.6^*	0.1	**0.2**	0.2	33.0^*	0.1^*	0.0^*	0.0^*	0.0^*	0.0^*
gauss_mix	5	**198.9**	37.5	44.2	48.7	18.5	47.0	3.9	1.2	0.5^*	10.8
gp_regr	3	**226.7**	45.0	43.6	43.2	94.6	93.0	31.7	31.9	4.9	14.4
earn_h	3	0.8^*	1.1	**14.5**	10.5	17.6^*	0.1^*	0.0^*	0.0^*	50.2^*	0.4^*
es_nc	10	8.1	16.8	15.9	18.1	3.5	**43.6**	10.6	11.6	1.8	14.0

classes are open problems, e.g., horseshoe and extreme skew). On the real posteriordb suite it takes 1 of 37 rows and fails the gate on 18, mostly low-dimensional raw-scale regressions. We flag the interpretation honestly: MCLMC's unadjusted dynamics carry an asymptotic bias that is a *tunable* trade-off [34], and its posteriordb gate failures conflate that intrinsic trade-off with our tuning choices (auto-tuned L and step size, no covariate standardization; per-baseline settings are listed in the protocol)—the safe conclusion is not that MCLMC cannot sample these posteriors but that an aggressively tuned configuration which excels on controlled families did not transfer, while PMR-HMC's single configuration did. *Third*, the four flow-based methods—TESS, flow-IMH, the flowMC-style hybrid, and the flow-preconditioned NeuTra, trained at the same budget our atlas warm-up spends—are not competitive on real posteriors: their dominant failure mode is accuracy (gate failures on 28–34 of 37 rows for the three independence-flow samplers), and on gate-passing rows PMR-HMC's median advantage is still 11–95×; among gate-passing rates none exceeds dense NUTS on that suite. A single global flow must represent the entire posterior geometry at once; the atlas replaces that hard learning problem with several easy chart fits plus an exact mixture, and the decomposition ablations below (Table 5) locate the gain: the atlas with exact harmonic dynamics carries most of it, the independence branch is decisive for multimodality, and neither branch alone is correct everywhere. On the multimodal mega-30 rows, all nine baselines fail the accuracy gate on every mixture target (mode-stuck: high ESS on the wrong distribution); PMR-HMC is the only method that passes.

9.5 Decomposition ablations: where the gain comes from

Three reduced kernels isolate the contributions, each run over 8 seeds with the accuracy gate applied: *zero-force* (atlas and exact harmonic dynamics, surrogate forced to zero), *global-only* (independence Metropolis–Hastings from the learned q alone, p_g=1), and the full kernel.

Table 3: posteriordb inclusion audit (46 gold-referenced posteriors, 37 ported; no posterior was excluded on account of results).

family	ported	reason if excluded
earnings regressions (7)	6/7	1 redundant standardized variant
kidiq (4)	4/4	—
kidiq_with_mom_work (4)	4/4	—
mesquite forms (6)	5/6	1 redundant un-logged variant
nes 1972–2000 (8)	8/8	—
blr (sblri, sblrc) (2)	2/2	—
eight_schools NC (1)	1/1	—
gauss mixture (1)	1/1	—
GP (gp_regr, gp_pois, accel_gp) (3)	1/3	latent-GP infrastructure
diamonds, kilpisjarvi (2)	2/2	—
time series (arK, arma, garch) (3)	3/3	—
HMM family (3)	0/3	forward-algorithm port
ODE (lotka–volterra, one_comp) (2)	0/2	ODE-solver port
total	37/46	

Table 4: Ten-method panel summary. Row 1–2: accuracy-gated best counts per suite (ties to the higher rate). Row 3–4: median PMR-HMC efficiency ratio over rows where *both* PMR-HMC and that baseline pass the gate (contested-row count in parentheses)—a gate-failing chain's ESS is not used as a denominator. Row 5–6: rows on which the baseline fails the gate. posteriordb: 37 posteriors, gold-judged; mega-30: 30 controlled targets, truth/reference-judged; 6 mega-30 rows (both funnels, the d=60 hierarchical Gaussian, Rosenbrock, banana at extreme bend, ν=2.5 Student-t) have no gate-passing method.

	PMR	NUTS	dNUTS	PF-dN	ChEES	MCLMC	TESS	f-IMH	flowMC	NeuTra
best (pdb)	**34**	0	2	0	0	1	0	0	0	0
best (mega)	**12**	1	1	0	0	9	1	0	0	0
ratio (pdb)	—	38× (35)	4.1× (35)	4.0× (35)	12× (11)	10× (19)	29× (9)	11× (7)	45× (3)	95× (31)
ratio (mega)	—	4.4× (16)	2.1× (17)	2.2× (17)	3.3× (8)	1.2× (15)	9.3× (9)	23× (8)	38× (7)	6.1× (15)
gate fails (pdb)	2	0	0	0	26	18	28	30	34	6
gate fails (mega)	9	11	11	11	21	13	20	22	22	13

Table 5 shows the pattern. On atlas-explained targets the independence branch with a strong q alone is fastest (mixture2: 497); off them it fails the gate outright (ring: 0.4 with 0.56σ error; banana: 1.5 log-variance error)—the local branch is what makes the kernel correct beyond the atlas's explanatory reach. The atlas plus exact harmonic dynamics carry most of the local gain: zero-force is within 1.4× of the full kernel on every target and ahead on two, with the cached surrogate contributing its clearest factor on the ring (45 → 64). The funnel row fails the variance gate for *every* kernel variant at this budget (as it does for all ten panel methods on the stress suite), and is reported for completeness, not claimed as a win. The nine baseline columns place the ablation in context: on these four canonical geometries every baseline fails the gate on the banana, the funnel, and the mixture, and the best gate-passing baseline on the ring (MCLMC at 3.3) sits 19× below the full kernel—even the *reduced* PMR kernels outperform every baseline wherever either passes. An earlier single-seed version of this table overstated the surrogate's contribution; these 8-seed medians supersede it.

Table 5: Decomposition ablations with the full baseline panel (min-ESS per 10^3 units; ablation columns and NUTS: medians over 8 seeds; other baselines: 4 seeds from the same-target modern-baseline battery; * fails the accuracy gate—a starred rate is not a valid win at any magnitude; **bold** = best gate-passing rate in the row). The funnel row is gate-failed by every kernel variant *and* every baseline.

target	full	zero-f.	glob.	NUTS	dNUTS	PF-dN	ChEES	MCLMC	TESS	f-IMH	flowMC	NeuTra
banana20	46.0	**52.9**	19.5*	0.5*	0.6*	0.5*	0.3*	1.6*	0.1*	0.8*	0.0*	0.5*
funnel10	2.3*	6.8*	0.6*	0.1*	0.1*	0.0*	0.2*	3.8*	0.2*	0.2*	0.2*	0.2*
mixture2	43.2	65.9	**497.0**	61.1*	65.5*	65.3*	93.8*	145.8*	23.0*	24.1*	3.4*	25.9*
ring20	**63.7**	45.4	0.4*	2.1	1.8	2.0	0.7*	3.3	0.2*	0.2*	0.0*	1.4

9.6 A deployed decision model at scale

On a faithful reproduction of a production two-stage control-function hurdle negative-binomial advertising model, seed-replicated over 4 seeds: median 21× at C=8 campaigns (d=37) and 11× at C=50 (d=163), with substantial seed variance at the larger size (per-seed rates 0.21–3.09 ESS per 10^3 units)—an earlier single-seed run showing 190× at C=50 was a favorable draw and is superseded by these medians. Both samplers degrade with C; in fact PMR-HMC's own median rate falls *more* (9.3×, driven by two poor warm-up seeds, versus 4.7× for NUTS), while remaining ≈ 11× more oracle-efficient at C=50; at C=120 (d=373) the current warm-up budget fails to establish acceptance (Section 10).

We additionally measured wall clock through the packaged library (Section 9.9; Python-callback density, one JIT-compiled evaluation per transition) at three campaign tiers. At C=30 (d=103): warm-up 81 s, production 12.4 s per 10^3 ESS at acceptance 0.14; at C=60 (d=193): 258 s and 47.6; at the largest tier C=120 (d=373) warm-up fails to establish acceptance (0.02), confirming the ceiling. Fully JIT-compiled NUTS on the same port is faster in wall clock at every tier (3.8/10.1/24.1 s per 10^3 ESS). At C=30 and C=60 the two samplers' first moments agree to within 0.25 posterior standard deviations on the recorded coordinates (both estimates carry sampling noise at these ESS); at C=120 PMR-HMC's ESS of 13 admits no meaningful agreement check. The reading is the crossover analysis of Section 9.9 in miniature: this port's likelihood compiles to microseconds, so gradients are effectively free and the oracle-cost separation does not convert into wall clock. For models of this kind the atlas kernel's wall-clock case requires an oracle-expensive deployment or heavy re-use of one fitted atlas, and the binding limitation at whale scale is the warm-up dimension ceiling, whose block-structured remedy we outline in Section 10.

9.7 A 30-target controlled stress panel

To probe the failure surface systematically under the full ten-method panel, we curated 30 targets from an 18-family parametric library (dimension 5..200, condition number to 10^5, correlation to 0.95, tail index ν = 2.5..10, curvature, mode separation, GLM size, hierarchy depth): every family represented, several at both an easy and a hard setting, with every *known PMR failure family deliberately retained* (funnels, hierarchical Gaussians, horseshoe, extreme skew, Rosenbrock, shell). Under the symmetric gate, PMR-HMC is gated-best on 12 of 30 rows, MCLMC on 9, and on 6 rows *no* method passes (both funnels, the d=60 hierarchical

Gaussian, Rosenbrock, banana at extreme bend, and ν=2.5 Student-t, where the variance gate is brutal for every sampler). The structure is informative. PMR-HMC's wins concentrate exactly where Theorem 4 operates: all three mixture rows (it is the *only* gate-passing method on any multimodal target), all correlated/ill-conditioned Gaussians (e.g. ρ=0.95: 191.0 vs. dense-NUTS 34.1; $\kappa=10^5$: 80.9 vs. 22.3), the easy banana and the ring (the hard banana is a no-pass row; the squiggle is a dead-heat tie awarded to TESS by rounding), and the diabetes and Poisson regressions. Its losses are the documented open chart-class problems—horseshoe and extreme skew (α=6) fail the gate outright, and on smooth exponential-tailed or GLM-type rows (log-cosh, logistic, probit, negative-binomial, Cauchy-noise, hierarchical binomial) MCLMC's tuned dynamics are 1.1–6.6× faster—plus one honest surprise: dense-mass NUTS edges PMR on the small production BCF variant (15.8 vs. 9.7 at d=49), though not at production scale (Section 9.6). The full 30×10 table with per-row gate flags is Table 8 (Appendix A); Section 9.8 tabulates every non-win row of both suites with its diagnosed mechanism.

9.8 Failure analysis: every row PMR-HMC does not win

Across both suites there are 21 rows on which PMR-HMC is not the gated-best method: 3 on posteriordb and 18 on mega-30. None is mysterious; each traces to one of four mechanisms, tabulated in Table 6 with the diagnosed root cause.

Each mechanism claim below was tested by a dedicated probe (Figure 10): production acceptance measured on every failing row, the triangular-chart fitter run on warm-up-style pool draws *versus* gate-passing reference draws, and the residual spread sd(R) evaluated on reference draws under the fitted atlas. The probes corrected two of our initial hypotheses, noted below.

(A) Representation or detection: the atlas family cannot express, or its detector cannot find, the structure. The shell target loses to MCLMC (3.6 vs. 8.1) because the polar chart unwraps a *planar* ring; a d-sphere shell concentrates on a norm level set and would need a radial chart (predicted fix, not yet tested). The extreme-bend banana (b=0.2, d=30; no method passes) is a measured expressiveness deficit: the fitted atlas leaves sd(R) = 1.07 nats on reference draws versus 0.52 on the easy banana it wins—and its acceptance is *healthy* (0.48), the signature of a chain that moves well inside a reference that misses curvature. The horseshoe belongs here too, which we learned from the probe: the triangular chart's log-square driver scan finds *no* driver even when handed 3,000 gate-passing reference draws (while firing spuriously on pool draws)—a *detector* gap, not a data gap, and our initial acquisition-only explanation was wrong.

(B) Acquisition: the chart class and detector work, but warm-up data never exposes the structure. The α=6 skew target is the proven case: the same fitter recovers strong skew (ε = 0.74) from reference draws but only $\varepsilon = 0.38$ from the warm-up pool, and the admitted chart is starved of forward-KL responsibility by the co-located Gaussian components (acceptance 0.027). Deep hierarchies (`hgauss60`: 15σ median error, acceptance 0.03) fail the same way, and the Rosenbrock valley is the purest instance: the shear fitter recovers the correct chart when handed oracle valley data, but warm-up exploration never collects such data (the success-history DE explorer ablation returned null, Section 9.11). The funnels and eight-schools sit between (B) and the budget ceiling, with degraded (not collapsed) acceptance 0.09–0.24.

The acceptance probe (Figure 10a) separates three deployment-observable signatures, none requiring reference draws: acquisition failures *collapse* (0.008–0.03 versus $\geq$ 0.88 on wins),

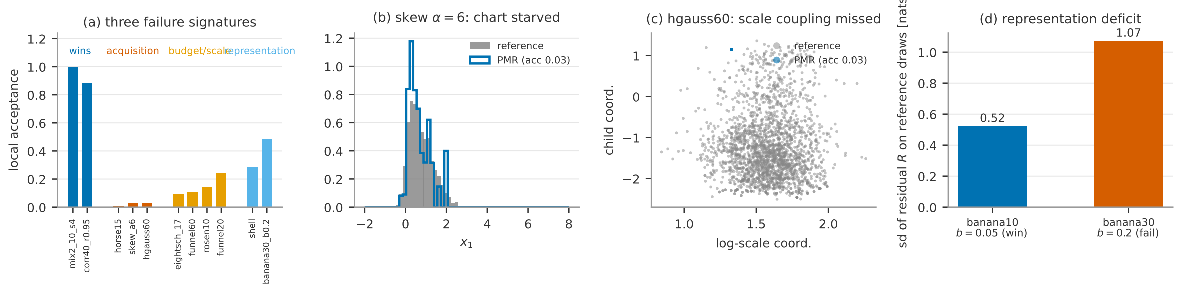

Figure 10: Failure-analysis probes. (a) Production acceptance separates three failure signatures: acquisition failures collapse (0.008–0.03), budget-limited rows degrade (0.09–0.24), representation failures keep healthy acceptance (0.29–0.48) while sampling a subtly wrong law; wins sit at 0.88–1.0. (b) The α=6 skew miss: PMR draws (blue) versus reference (gray)—the admitted triangular chart is starved and under-corrects the skew. (c) The d=60 hierarchical Gaussian: the chain never tracks the scale–child coupling. (d) The extreme-bend banana's measured expressiveness deficit: residual spread on reference draws doubles relative to the winning easy-banana row.

budget-limited rows degrade (0.09–0.24), and representation failures keep healthy acceptance while the chain samples a subtly wrong law—the one signature that instead requires the cross-seed rank-$\widehat{R}$ diagnostic, which catches both such cases on posteriordb (1.11, 1.26).

(C) No-advantage regime: the geometry is already easy for gradient samplers. On hand-noncentered eight-schools (posteriordb), smooth well-scaled GLM rows (logistic, probit, negative-binomial, Cauchy-noise, log-cosh, hierarchical binomial), and the small production-BCF variant, PMR-HMC *passes* the accuracy gate but a tuned gradient sampler is 1.1–6.6$\times$ faster per oracle unit: when the posterior is close to an isotropic Gaussian after trivial rescaling, an atlas has nothing to explain and its warm-up is pure overhead. These are efficiency losses, not correctness failures.

(D) Reference-limited rows. `earn_height` (extreme heteroscedastic raw-scale tail) and `kilpisjarvi` are genuine atlas misfits—and both are flagged *without gold draws* by cross-seed rank-$\widehat{R}$ (1.11, 1.26); on ν=2.5 Student-t the population variance barely exists and every method fails the variance gate, a limit of the gate itself as much as of any sampler.

The remedies map one-to-one: (A) a radial chart, richer shear elements, and—for the horseshoe—a shrinkage-aware driver detector (the inverse-KL triangular objective of Section 10 subsumes the current scan); (B) visited-state refits and population-based acquisition (Section 10); (C) is inherent—the correct reading is that PMR-HMC's advantage is a function of how much structure the posterior has for an atlas to remove; (D) is shared by all methods.

9.9 Wall clock: native kernel, both regimes

A $\sim$700-line C kernel consumes the frozen atlas (all chart classes, both cache types, winding-branch and Gamma-auxiliary sampling) at 0.13–385 μs per draw, with target-formula parity $\leq 10^{-12}$ against the JAX implementation on the 13-target parity study ($\leq 7.3 \times 10^{-12}$ across all 66 native-family targets) and statistically indistinguishable acceptance. We report both wall-clock regimes honestly, on the 66 parametric-suite targets with native C density implementations (4 seeds, identical double precision). *Production rate*: the C kernel attains higher

Table 6: Every non-win row, diagnosed; the acc. column is the measured production acceptance from the verification probe. Class: A = missing chart class *or* detector, B = warm-up acquisition, C = no-advantage regime (gate-passing efficiency loss), D = reference-limited. “none” = no method passes the gate on that row.

row(s)	winner	class	acc.	diagnosed cause (probe-verified)
shell	MCLMC	A	.29	polar chart is planar; radial chart is the predicted fix
banana b=0.2, d=30	none	A	.48	expressiveness: sd(R) 1.07 vs 0.52 nats on the win
horseshoe (d=15)	MCLMC	A	.008	driver scan finds nothing even on reference draws
skew α=6	MCLMC	B	.027	fitter recovers ε=0.74 from reference, 0.38 from pool
hgauss60	none	B	.03	scale coupling unexposed; 15σ error
funnels, 8-schools	none/NUTS	B	.09–.24	degraded at budget; between (B) and the d ceiling
Rosenbrock d=10	none	B	.14	valley data never acquired (fitter itself correct)
GLM rows (×6)	MCLMC	C	—	near-Gaussian after rescaling; 1.1–6.6×
eight-schools NC (pdb)	MCLMC	C	—	hand-noncentered; atlas has nothing to remove
production BCF d=49	dNUTS	C	—	dense mass suffices (hypothesis, not probed)
squiggle f=4	TESS	C	—	dead heat (15.3 vs 15.3), tie by rounding
earn_height	dNUTS	D	—	heteroscedastic raw-scale tail; $\widehat{R}$ 1.11
kilpisjarvi	dNUTS	D	—	all methods $\leq$ 0.6; $\widehat{R}$ 1.26
Student-t ν=2.5	none	D	—	variance gate unattainable for every sampler

ESS/s than fully JIT-compiled NUTS on 54 of 66 targets, median 13×, exceeding 10^2–10^5× on correlated, ill-conditioned, and multimodal targets. *End-to-end at a small budget*: charging PMR-HMC’s Python warm-up against a request of only 1,000 effective samples, NUTS is faster on the majority of targets (PMR-HMC wins 17 of 66)—at tiny budgets the warm-up dominates and a JIT-compiled gradient sampler is hard to beat on easy targets. The crossover budget—the requested ESS beyond which PMR-HMC wins end-to-end—has median ≈ 4,500 ESS (interquartile 730–11,400) over the 54 production-faster targets. This is precisely the amortization analysis of Section 8.4 realized in wall clock: single short chains on easy targets favor NUTS; multi-chain runs, large ESS budgets, repeated re-use of a fitted atlas (the production setting of Section 9.6), and any target hard enough that NUTS’s per-ESS cost blows up all sit far past the crossover.

9.10 Calibration as an implementation audit

Exactness follows from the invariant-kernel construction (Section 5); simulation-based calibration provides an end-to-end *implementation* check. Over three prior-predictive families × 30 replications, rank statistics are uniform (worst family: 1 of 8 parameters below p=0.05, consistent with chance at the 5% level, where 0.4 rejections among 8 tests are expected). Scaling SBC to several hundred replications per family is planned for the revision.

9.11 Further ablations and negative results

A Pareto/CMA-ES environmental-selection atlas builder [7, 11] attains parity with the score rule while producing leaner atlases (funnel K=7 vs. 11); a success-history DE explorer [47] for warm-up acquisition produced a null result (2 seeds × 4 targets), recorded to delimit what does not yet solve valley acquisition. Removing the provenance correction (26) collapses atlases (importance ESS → 1); replacing branch sampling (15) by a principal branch measurably breaks calibration.

10 Limitations

(1) Hand-noncentered models are the gradient samplers' best case; our margin there is small or negative (MCLMC takes noncentered eight-schools). (2) Products of coupled funnels (horseshoe priors) defeat the current warm-up: the triangular chart could express them, but the pool never exhibits the per-coordinate scale variation its driver scan needs. (3) Warm-up has a dimension ceiling near $d \approx 400$ under fixed budgets; block-structured atlas construction is the natural remedy. (4) Valley data acquisition (Rosenbrock-class) is solved at the chart level but not yet at the warm-up-exploration level. (5) The warm-up gradient bill is $\Theta(\sqrt{\kappa})$, paid once. (6) ODE, HMM, and latent-GP posterior families await ports (Table 3). (7) All guarantees are for the frozen kernel; online atlas refinement requires a cross-fitted ensemble construction left to future work. (8) Uniform ergodicity is conditional on tail dominance (Theorem 2); the defensive component makes the condition plausible, not automatic. (9) Strong skewness remains open at the acquisition level. The sinh-arcsinh triangular chart of Section 6—a restricted Knothe–Rosenblatt form in the spirit of adaptive transport maps [25, 32]—already closes the exponential-tail gap (log-cosh now passes the gate) and helps moderate skew, but at α=6 skew and on the horseshoe the fitted chart is starved: forward-KL responsibilities concentrate on the Gaussian components, and the L-BFGS-anchored warm-up pool does not expose the driver variation the fit needs. The remedies are richer triangular elements fitted by the convex, coordinate-separable inverse-KL objective (whose first-order conditions are Stein-type generalized moment conditions subsuming our hand-crafted detectors), refitting transports from visited states across warm-up rounds, and the acquisition improvements of (4). The winding chart survives on topology grounds.

11 Conclusion

Removing learned transport structure first, and approximating only the residual, converts a broad class of practitioner posteriors into near-independent sampling at one target-density evaluation per draw. The construction is exact by design at every extension point—arbitrary cached forces, winding charts, heavy-tailed charts—realizes measured complexity separations against strong modern baselines, and compiles to a kernel fast enough that the oracle, rather than the sampler, is again the dominant cost of Bayesian computation.

A Complete Ten-Method Panel Results

Both tables report median min-ESS per 10^3 target-oracle units (density = 1 unit, gradient = 2.5); * marks an accuracy-gate violation (max standardized mean error $> 0.25\sigma$ or max $|\log|$-variance error > 0.5 against gold draws or analytic truth/reference moments); **bold** marks the best rate among gate-passing methods in the row; "–" marks a method that errored. PMR/NUTS/dense-NUTS medians are over 8 seeds on posteriordb and 4 on mega-30; all other methods over 4 seeds.

Table 7: posteriordb: 37 posteriors × 10 methods, gold-judged.

target	PMR	NUTS	dNUTS	PF-dN	ChEES	MCLMC	TESS	f-IMH	flowMC	NeuTra
pdb_arK	**147.0**	4.9	47.3	47.8	11.9	9.4	3.9	3.7	0.1*	15.3
pdb_arma11	**170.5**	40.2	43.3	42.2	21.4	55.1	1.2	0.0*	25.2*	9.6
pdb_diamonds	**63.2**	0.1	14.7	10.1	0.0*	0.1*	0.0*	0.0*	0.0*	0.0
pdb_earn_h	0.8*	1.1	**14.5**	10.5	17.6*	0.1*	0.0*	0.0*	50.2*	0.4*
pdb_es_nc	8.1	16.8	15.9	18.1	3.5	**43.6**	10.6	11.6	1.8	14.0
pdb_garch11	**63.6**	11.6	17.7	17.8	4.7	24.7	8.9	6.0	1.4	9.6
pdb_gauss_mix	**198.9**	37.5	44.2	48.7	18.5	47.0	3.9	1.2	0.5*	10.8
pdb_gp_regr	**226.7**	45.0	43.6	43.2	94.6	93.0	31.7	31.9	4.9	14.4
pdb_kid_hs	**84.4**	9.6	37.6	33.1	96.3*	0.5*	0.0*	0.0*	0.0*	2.5
pdb_kid_iq	**228.1**	2.8	25.9	17.1	18.4*	0.2*	0.0*	0.0*	0.0*	1.6
pdb_kid_ix	**155.5**	0.6	20.4	15.3	15.2*	0.1*	0.0*	0.0*	0.1*	0.1*
pdb_kid_ixc	**165.6**	32.2	38.8	36.2	64.0*	0.3*	0.0*	0.0*	0.0*	0.6
pdb_kid_ixc2	**191.5**	14.0	43.2	38.6	64.5*	0.3*	0.0*	0.0*	0.0*	0.3
pdb_kid_ixz	**183.1**	25.9	32.0	25.7	65.6*	0.3*	0.0*	0.0*	0.1*	0.0*
pdb_kid_work	**175.1**	5.3	28.7	23.9	90.7*	0.3*	0.0*	0.0*	0.0*	0.1*
pdb_kidiq	**199.1**	2.6	19.4	12.5	0.0*	0.1*	0.0*	0.0*	0.0*	0.5
pdb_kilpis	0.6*	0.1	**0.2**	0.2	33.0*	0.1*	0.0*	0.0*	0.0*	0.0*
pdb_log10earn	**214.5**	1.2	1.9	1.4	23.1*	0.1*	0.0*	0.0*	0.0*	1.3
pdb_logearn	**239.0**	1.1	4.1	2.6	19.2*	0.1*	0.0*	0.0*	50.1*	0.8
pdb_logearn_hm	**219.1**	0.8	4.5	3.2	12.7*	0.1*	0.0*	0.0*	0.0*	0.5
pdb_logearn_ix	**193.6**	0.4	3.7	2.6	6.4*	0.1*	0.0*	0.0*	0.0*	0.1
pdb_logearn_lhm	**206.3**	0.2	23.7	21.9	0.0*	0.1*	0.3*	0.2*	25.1*	0.1*
pdb_mesq_full	**59.7**	3.6	36.2	40.0	2.7	11.0	2.0	0.3*	50.2*	3.3
pdb_mesq_va	**98.5**	3.7	40.8	41.1	3.1	9.6	3.0	2.7	50.2*	10.1
pdb_mesq_vas	**56.4**	2.1	38.7	35.7	1.0*	6.1	0.8*	0.1*	50.2*	1.4
pdb_mesq_vash	**54.4**	2.0	35.0	39.3	1.2	7.0	0.7*	0.3*	50.2*	1.4
pdb_mesquite	**156.4**	22.5	38.0	38.9	49.4	53.3	15.9	15.8	50.2*	11.5
pdb_nes1972	**142.2**	3.1	43.2	45.7	0.4*	5.6	0.0*	0.0*	0.0*	1.8
pdb_nes1976	**150.8**	3.0	47.3	43.1	0.8*	6.1	0.0*	0.0*	0.0*	1.5
pdb_nes1980	**148.4**	2.8	49.9	45.2	0.5*	7.3	0.0*	0.0*	0.0*	1.4
pdb_nes1984	**143.7**	2.9	44.2	43.6	0.7	6.4	0.0*	0.0*	0.0*	1.5
pdb_nes1988	**143.2**	2.7	46.7	46.1	0.7*	5.7	0.0*	0.0*	0.0*	1.7
pdb_nes1992	**144.7**	2.9	48.5	43.1	0.8*	6.8	0.0*	0.0*	0.0*	1.7
pdb_nes1996	**145.9**	2.5	49.8	46.4	0.5*	8.5	0.0*	0.0*	0.0*	1.2
pdb_nes2000	**132.9**	2.3	45.5	45.3	0.3*	5.5	0.0*	0.0*	0.0*	1.2
pdb_sblrc	**121.7**	3.2	3.3	2.6	1.0*	1.3*	0.0*	0.0*	0.0*	0.9
pdb_sblri	**122.9**	6.4	6.9	5.3	2.0*	3.3*	0.0*	0.0*	0.0*	0.4

Table 8: mega-30 controlled stress panel: 30 targets $\times$ 10 methods, truth/reference-judged.

target	PMR	NUTS	dNUTS	PF-dN	ChEES	MCLMC	TESS	f-IMH	flowMC	NeuTra
m_banana10_b0.05	**95.5**	0.2	0.7*	1.3*	0.9*	4.1	0.3*	1.7*	0.1*	0.4*
m_banana30_b0.2	6.5*	0.4*	0.3*	0.3*	0.2*	0.4*	0.0*	0.8*	0.0*	0.1*
m_cauchyreg15_43	61.3	33.2	50.8	49.5	10.9	**91.1**	9.2	6.1	2.7	18.7
m_corr150_r0.9	**89.0**	0.1*	3.0	2.9	0.3*	1.2	0.1*	0.0*	0.0*	0.0
m_corr40_r0.95	**191.0**	0.2*	34.1	29.0	0.3*	2.2	0.2*	0.2*	0.0*	0.2
m_diab_19	**124.4**	2.2	62.2	60.5	1.5	7.2	0.7*	0.1*	0.1*	3.5
m_eightsch_17	1.7*	**0.2**	0.3*	0.0*	0.1*	1.1*	0.0*	0.1*	0.1*	0.1*
m_funnel20	7.9*	0.0*	0.0*	0.0*	0.1*	2.7*	0.1*	0.3*	0.0*	0.1*
m_funnel60	0.2*	0.0*	0.0*	0.0*	0.2*	0.7*	0.0*	0.1*	0.0*	0.0*
m_gauss_iid100	**159.9**	26.5	2.1	2.3	25.4	127.3	10.6	3.4	0.9	10.8
m_gauss_ill200_k1e4	**51.9**	0.4	0.6	1.0	0.1*	0.1*	0.0*	28.4*	50.2*	0.0*
m_gauss_ill20_k1e+05	**80.9**	0.1	22.3	20.1	0.0*	0.1*	0.0*	0.0*	25.1*	0.0*
m_hbinom75	20.6	20.5	8.3	9.2	2.7*	**72.4**	2.4	0.8*	0.1*	9.6
m_hgauss60	0.1*	0.0*	0.0*	0.0*	0.1*	0.9*	0.0*	0.0*	0.0*	0.1*
m_horse15	0.0*	3.6	4.3	4.0	0.1*	**7.0**	0.1*	0.0*	0.0*	3.2
m_logcosh25	10.6	20.8	14.9	13.3	25.8	**69.5**	12.9	2.9	0.3*	11.8
m_logreg30_12	26.7	11.6	27.6	24.8	17.4	**55.9**	2.9	1.1	0.7	8.7
m_mix2_10_s4	**43.8**	0.0*	0.1*	57.7*	86.4*	0.1*	23.2*	24.5*	2.0*	0.0*
m_mix2_30_s8	**39.3**	55.1*	21.7*	19.6*	91.2*	121.2*	15.3*	11.5*	1.5*	26.2*
m_mix3	**34.4**	0.2*	0.4*	1.0*	93.8*	0.8*	12.7*	15.5*	1.9*	0.1*
m_negbin25_39	81.1	42.8	38.7	39.8	23.1	**95.2**	6.7	3.9	1.6	17.9
m_pois25_23	**91.5**	33.2	51.7	44.4	53.1	76.1	7.0	1.8	2.4	15.1
m_probit30_34	83.7	33.0	40.9	33.8	26.6	**91.7**	8.8	2.0	1.5	16.7
m_ring_R12	**66.6**	1.6	1.4	1.7	0.9*	2.9	0.0*	0.1*	0.0*	1.3
m_rosen10	0.2*	0.3*	0.2*	0.2*	0.1*	0.4*	0.0*	0.0*	0.0*	0.1*
m_shell	3.6	3.9	3.0	3.3	0.4*	**8.1**	0.0*	0.2*	0.1*	2.6
m_skew_a6	0.6*	12.6	13.0	12.1	15.5	**29.4**	1.1	0.3*	0.4	7.4
m_squig_f4	15.3	0.2	0.2	0.2	0.5*	0.3*	**15.3**	7.6	1.8	0.2
m_t30_nu2.5	29.7*	29.1*	10.0*	11.6*	12.2*	9.4*	6.3*	0.1*	0.3*	13.0*
m_vilya12	9.7	0.6	**15.8**	13.8	0.1*	2.4	0.0*	0.0*	0.0*	0.1

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